

LEGEND								
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	
AD	ACCESS DOOR		SECTION DESIGNATION: UPPER—SECTION IDENTIFICATION LOWER—NUMBER OF SHEET ON WHICH SECTION IS DRAWN		FLOW METER		DETAIL DESIGNATION	
AP	ACCESS PANEL				FLEXIBLE DUCTWORK		UPPER — DETAIL DESIGNATION NUMBER LOWER LEFT — DWG CONDITION IS SHOWN LOWER RIGHT — DWG DETAIL IS SHOWN	
AS	AIR SUPPLY		UPPER—SECTION IDENTIFICATION LOWER—NUMBER OF SHEET ON WHICH SECTION IS DRAWN		ACOUSTICAL DUCT LINER		DESIGNATION MAX. CFM MIN. CFM	VARIABLE AIR VOLUME BOX
AFF	ABOVE FINISHED FLOOR				TURNING VANES		DESIGNATION MAX. CFM PRIMARY CFM MIN. CFM	FAN POWERED CONSTANT VOLUME BOX
BTUH	BRITISH THERMAL UNIT PER HOUR		SMOKE DETECTOR		FIRE DAMPER			TWO-WAY MODULATING CONTROL VALVE
CMCS	COMPUTER MONITORING AND CONTROL SYSTEM		REFRIGERANT LEAK DETECTOR		BACKDRAFT DAMPER			SAFETY VALVE OR PRESSURE RELIEF
CFM	CUBIC FEET PER MINUTE		THERMOSTAT		VOLUME DAMPER			AUTOMATIC AIR VENT OR VACUUM
FC	FLEX. CONN.		HUMIDISTAT		MOTORIZED DAMPER			MANUAL AIR VENT
DB	DRY BULB TEMPERATURE, DEG. F.		WATER PRESSURE GAUGE		OPPOSED BLADE DAMPER(AUTOMATIC)			ANGLE GATE VALVE
EAT	ENTERING AIR TEMPERATURE		AIR CONDITIONING UNIT		PNEUMATIC DAMPER (AUTOMATIC)			ANGLE GLOBE VALVE
EDB	ENTERING DRY BULB		CONDENSING UNIT		HIGHW PRESSURE STEAM SUPPLY			THREE-WAY MODULATING CONTROL VALVE
ENT.	ENTERING		AIR HANDLING UNIT		HIGH PRESSURE STEAM RETURN			CONCENTRIC REDUCER
EWB	ENTERING WET BULB		AIR SEPARATOR		LOW PRESSURE STEAM SUPPLY			UNION OR FLANGE
EWT	ENTERING WATER TEMPERATURE		BOILER		LOW PRESSURE STEAM RETURN			CAPPED PIPE
FD	FIRE DAMPER		CABINET HEATER		HEAT RECOVERY (GLYCOL)			DIRECTION OF FLOW
FPM	FEET PER MINUTE		FAN COIL UNIT		PUMPED CONDENSATE			STRAINER (WITH 3/4 BLOW-OFF VALVE)
WTR.	WATER		CABINET UNIT HEATER		BOILER FEED			GUIDE
GPM	GALLONS PER MINUTE		EXHAUST FAN		BLOWDOWN			ANCHOR
HP	HORSEPOWER		EXPANSION TANK		REFRIGERANT LIQUID			PIPE RISER
FE	FLOW ELEMENT		FINNED TUBE RADIATION		REFRIGERANT SUCTION			PIPE DROP
LWT	LEAVING WATER TEMPERATURE		FLOW METER		CHILLED WATER SUPPLY			FLEXIBLE PIPE CONNECTION
MAX.	MAXIMUM		PUMP		CHILLED WATER RETURN			FLOW SWITCH
MIN.	MINIMUM		SOUND ATTENUATOR		VENT LINE			THERMOMETER
NC	NORMALLY CLOSED		UNIT HEATER		COLD (MAKEUP) WATER			EXPANSION JOINT
N.I.C.	NOT IN CONTRACT		FAN POWERED CONST. VOL. BOX		GATE VALVE			BALL VALVE
NO	NORMALLY OPEN		UPPER-EQUIPMENT DESIGNATION		GLOBE VALVE			PLUG VALVE
NOM.	NOMINAL		LOWER-EQUIPMENT NUMBER		CHECK VALVE			VALVE WITH HOSE BIBB
OA	OUTSIDE AIR		HEATING & VENTILATING UNIT		BALANCING VALVE			PUMP
OBD	OPPOSED BLADE DAMPER		HEAT PUMP		COMBINATION BALANCING AND SHUT-OFF VALVE			NON-SLAM CHECK VALVE
LWT	LEAVING WATER TEMPERATURE		REHEAT COIL		PRESSURE REDUCING VALVE OR BACK PRESSURE CONTROL VALVE			BUTTERFLY VALVE
PD	PRESSURE DROP		HEAT EXCHANGER		FLOW MEASURING DEVICE			SOLENOID VALVE
LDB	LEAVING DRY BULB		MOTOR		TYPE NECK SIZE CFM			LEVER OPERATED, SPRING LOADED VALVE W/FUSIBLE LINK
PRV	PRESSURE REDUCING VALVE		SUPPLY DUCT UP		BALL VALVE			4 1/2" PRESURE GAGE WITH PIGTAIL AND PETCOCK
PSIG	POUNDS PER SQUARE INCH GAUGE		SUPPLY DUCT DOWN		POINT OF CONNECTION NEW TO EXISTING			STEAM TRAP (TYPE AS INDICATED)
RA	RETURN AIR		EXHAUST OR RETURN DUCT UP					
RPM	REVOLUTIONS PER MINUTE		EXHAUST OR RETURN DUCT DOWN					
HW	HOT WATER		ROUND DUCT UP					
SP	STATIC PRESSURE (INCHES OF WATER)		ROUND DUCT DOWN					
WB	WET BULB TEMPERATURE, DEG. F.		WIRE SCREEN					
LAT	LEAVING AIR TEMPERATURE		FLEXIBLE DUCT CONNECTION					
MBH	BTUH (THOUSANDS)		SOUND ATTENUATOR					

REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	AS-BUILT, NO CHANGES	J.P.	9/24/99	R.R.

GENERAL NOTES

- ALL PIPING AND EQUIPMENT SHOWN DIAGRAMATICALLY ONLY. GENERAL ORIENTATION SHOWN IN PLAN AND SECTIONAL DRAWINGS. EXACT LOCATIONS TO BE DETERMINED IN FIELD.
- INSTALL PIPING STRAIGHT, PLUMB AND AS DIRECT FORM RIGHT ANGLES ON LINES PARALLEL WITH BUILDING WALLS.
- INSTALL PIPES CLOSE TO WALLS, PARTITIONS, CEILINGS, ETC. OFFSET ONLY WHERE NECESSARY TO FOLLOW WALLS, AS INDICATED.
- LOCATE GROUPS OF PIPES PARALLEL TO EACH OTHER; SPACE THEM TO PERMIT APPLICATION OF FULL INSULATION AND TO PERMIT ACCESS FOR SERVICING VALVES.
- MANUFACTURERS' PRINTED INSTALLATION DIRECTIONS SHALL BE FOLLOWED.
- ALL PIPING SHALL BE REAMED TO BE FREE OF BURRS. KEEP PIPING FREE FROM SCALE AND DIRT; PROTECT OPEN PIPE ENDS WHENEVER WORK IS SUSPENDED DURING CONSTRUCTION TO PREVENT FOREIGN MATERIAL ENTERING AND LODGING; USE TEMPORARY PLUG, BURLAP, OR OTHER APPROVED MATERIAL FOR PROTECTION.
- UNIONS, SCREWED OR FLANGED, SHALL BE PROVIDED WHERE INDICATED AND IN THE FOLLOWING LOCATIONS, EVEN IF NOT INDICATED:
 - IN LONG RUNS OF PIPING TO PERMIT CONVENIENT DISASSEMBLY FOR ALTERATIONS OR REPAIRS.
 - IN BY-PASSES AROUND EQUIPMENT.
 - IN CONNECTIONS TO TANKS, PUMPS, OTHER EQUIPMENT REQUIRING DISCONNECTION FOR REPAIRS AND/OR REPLACEMENT; LOCATE BETWEEN SHUT-OFF AND EQUIPMENT.
 - ON BOTH SIDES OF TRAPS, CONTROLS AND AUTOMATIC CONTROL VALVES.
- WHERE STEEL FLANGES ARE MADE UP TO CAST IRON FLANGES, THE RAISED FACE ON THE STEEL FLANGE SHALL BE REMOVED. ALL FLAT FACED FLANGED JOINTS SHALL HAVE FULL-FACED GASKETS.
- PROVIDE SHUT-OFF VALVES WHERE INDICATED AND SPECIFIED, AND IN THE FOLLOWING LOCATIONS, EVEN IF NOT INDICATED OR SPECIFIED:
 - INDIVIDUAL EQUIPMENT UNITS AT INLETS AND OUTLETS TO PERMIT UNIT REMOVAL FOR REPAIRS WITHOUT INTERFERING WITH REMAINDER OF SYSTEM.
 - VALVES SHALL BE LOCATED FOR EASY ACCESS AND OPERATION.
 - VALVES SHALL NOT BE LOCATED WITH STEMS BELOW THE HORIZONTAL.
- PIPING SHALL BE INSTALLED TO ALLOW FOR EXPANSION, USING OFFSETS, SWING JOINTS, GUIDES AND ANCHORS, ETC. TO PREVENT UNDUE STRAIN ON PIPING OR EQUIPMENT.
- THESE ARE STANDARD SYMBOLS AND MAY NOT APPEAR ON THE PROJECT DRAWINGS. HOWEVER, WHEREVER THE SYMBOL ON THE PROJECT DRAWINGS OCCURS, THE ITEM SHALL BE PROVIDED AND INSTALLED.
- FOR SF-1, EF-5, EF-6, EF-8, EF-9, & EF-16 CONNECT FANS TO LOUVER WITH 12" MIN. DEEP PLENUM (INSULATED).

graphic scales

CHECK GRAPHIC SCALES BEFORE USING

ARROWSTREET INC
SOMERVILLE, MA

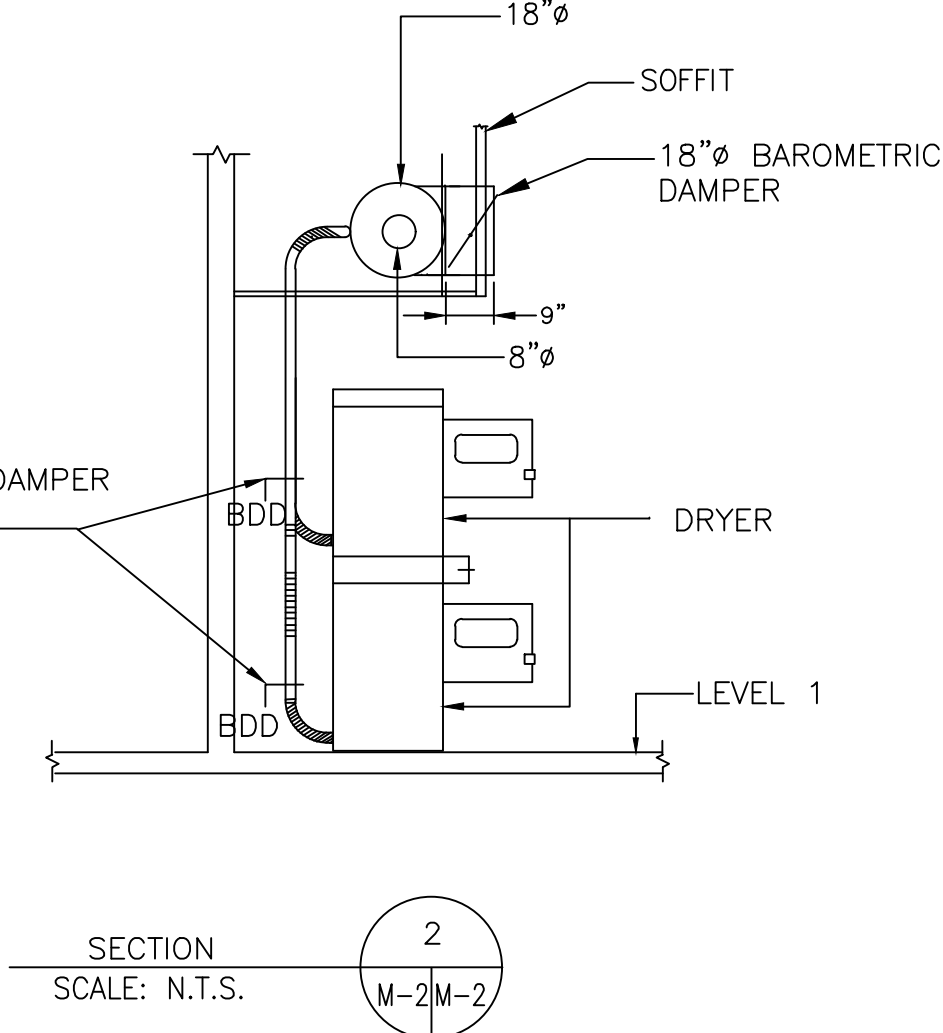
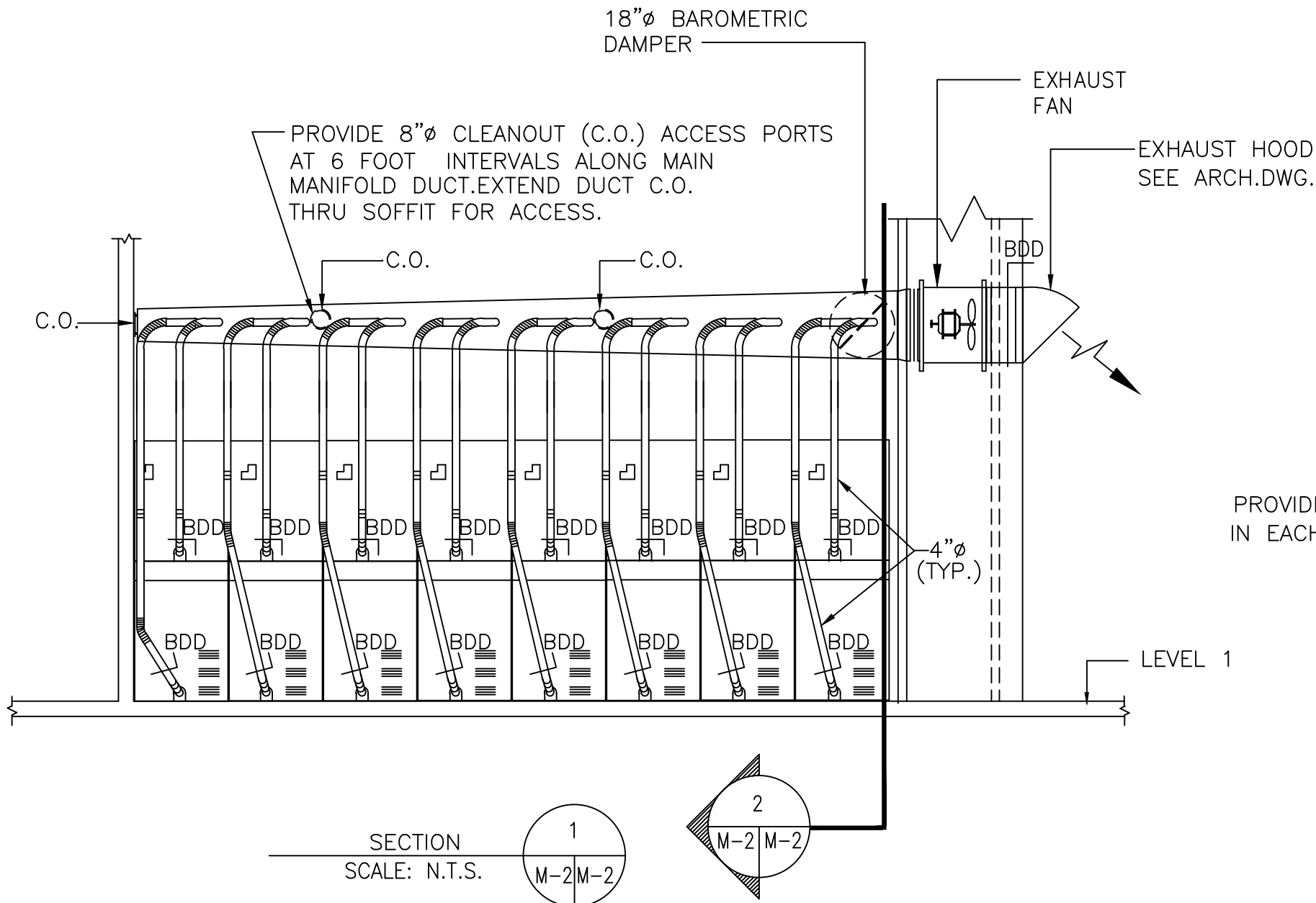
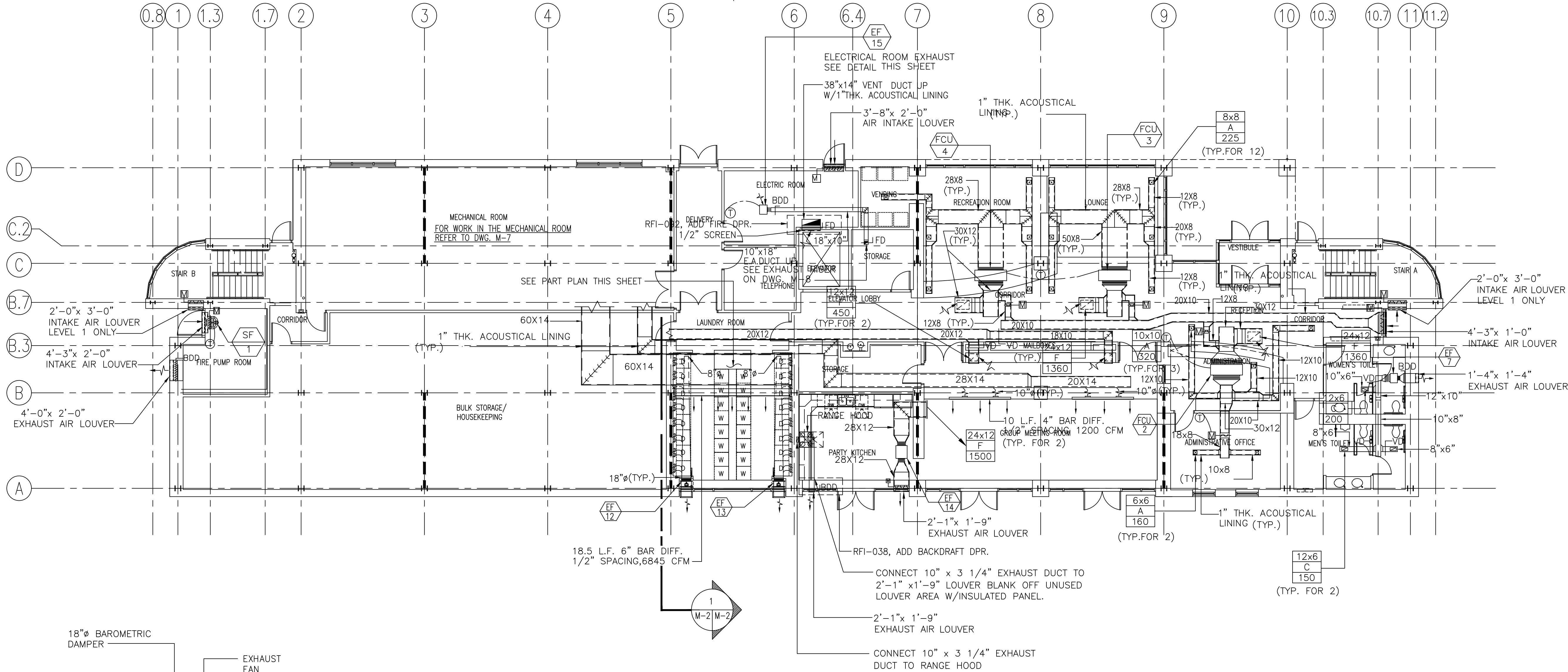
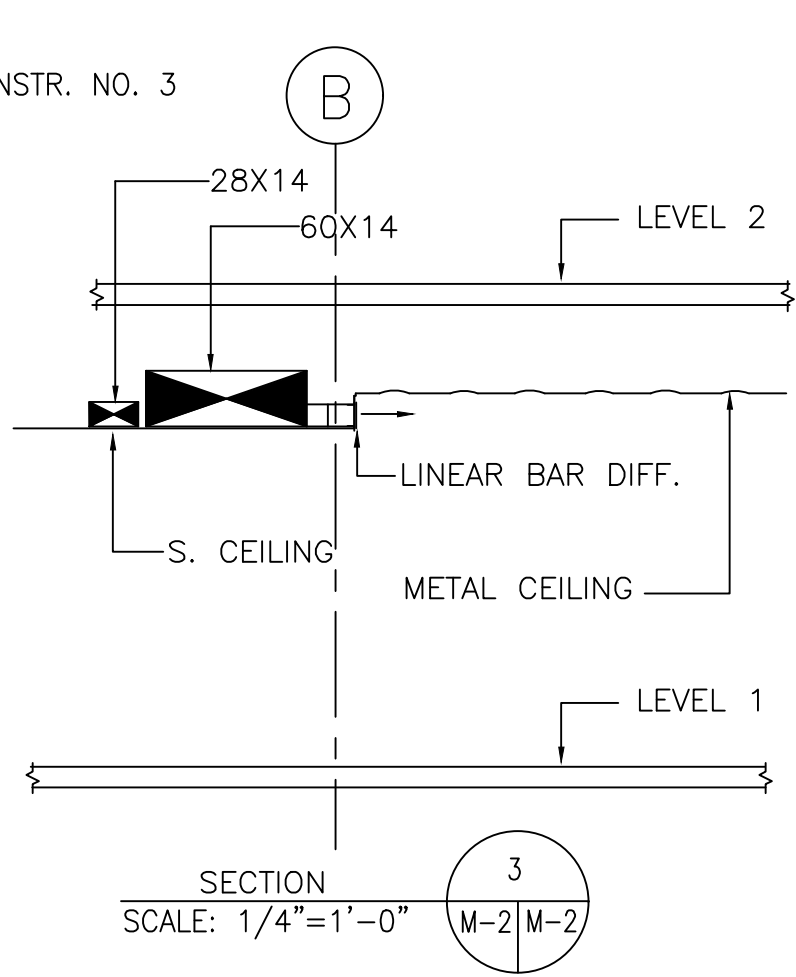
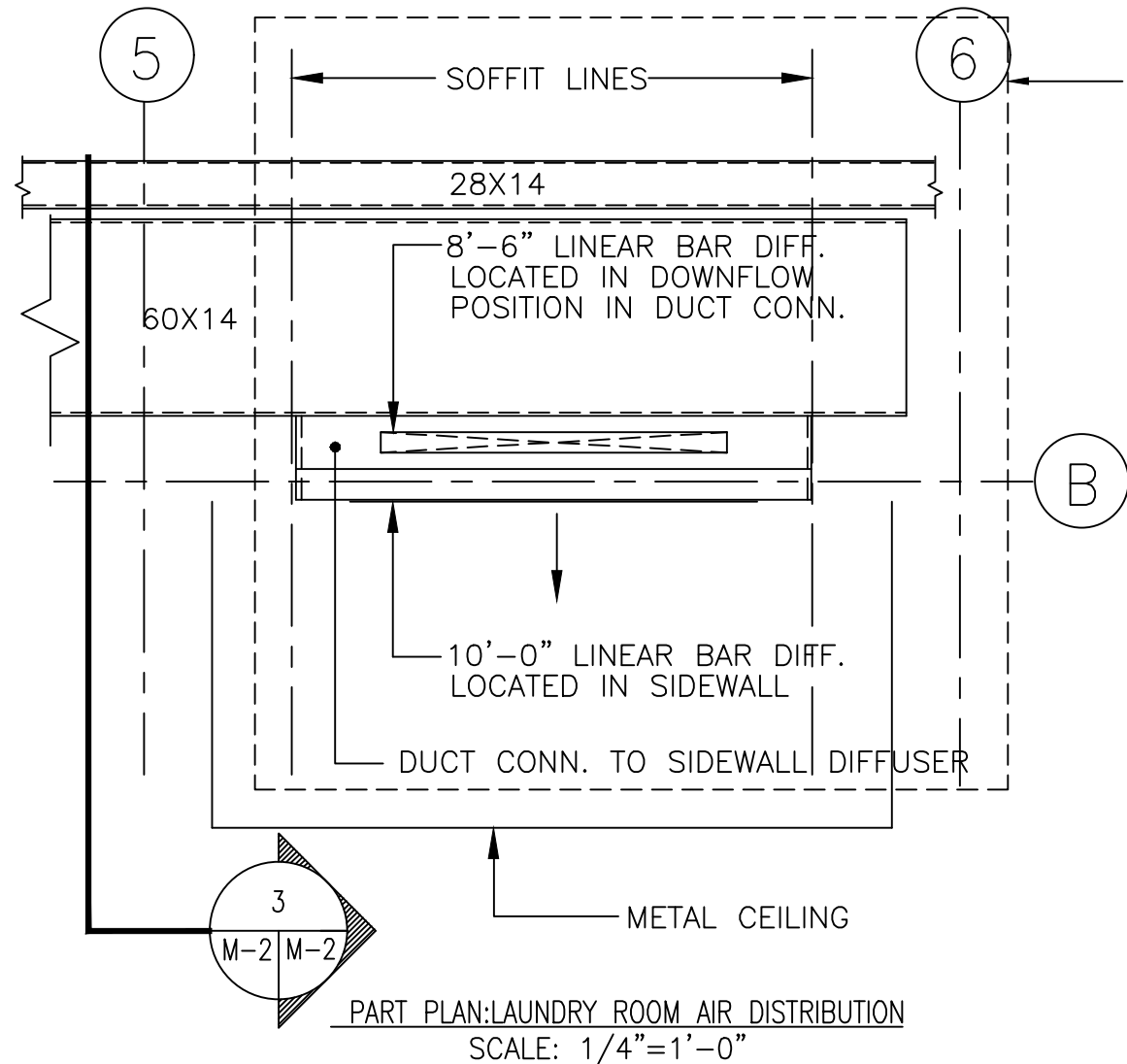
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SUPV RR CHK ENGR RR
SUBMITTED BY: W. DANIELL / PRINCIPAL DATE: 28 MARCH 1997
FIRM MEMBER (TITLE)
NORTHDIV HD
DM HD
PPE DRG
APPROVED DATE
OFFICER IN CHARGE DATE
NORTHDIV FOR COMMANDER NAVFAC

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
LESTER
NAVAL SUBMARINE BASE-NEW LONDON
PENNSYLVANIA
GROTON, CT

NORTHERN DIVISION
BACHELOR ENLISTED
QUARTERS
LEGENDS & ABBREVIATIONS

SIZE: F
CODE IDENT. NO.: P-422
NAVFAC DRAWING NO.:
DIS. SH. NO.: M-1
CONSTR. CONTR. NO. N62472-96-C-0420
SPEC. 04-96-0420 SHEET 80 OF 135

REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	CHANGED TO AS - BUILT	J.P.	9/24/99	R.R.



GROUND FLOOR PLAN
SCALE: 1/8"=1'-0"

GRAPHIC SCALES

CHECK GRAPHIC SCALES BEFORE USING

0 2' 4' 8'

SCALE: 1/4" = 1'

graphic scales

CHECK GRAPHIC SCALES BEFORE USING

0 4' 8' 16'

SCALE: 1/8" = 1'

ARROWSTREET INC
SOMERVILLE, MA

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NAVAL SUBMARINE BASE-NEW LONDON
PENNSYLVANIA
GROTON, CT

NORTHERN DIVISION
BACHELOR ENLISTED
QUARTERS

GROUND FLOOR - DUCTWORK PLAN

DSGN RDL DR RDL CHK RR
SUPV RR CHK ENGR RR

SUBMITTED BY: W. DANIEL / PRINCIPAL DATE: 28 MARCH 1997

FIRM MEMBER (TITLE)

NORTH DIV HD

DM HD

PPE DR

APPROVED DATE

OFFICER IN CHARGE DATE

APPROVED DATE

NORTH DIV FOR COMMANDER NAVFAC

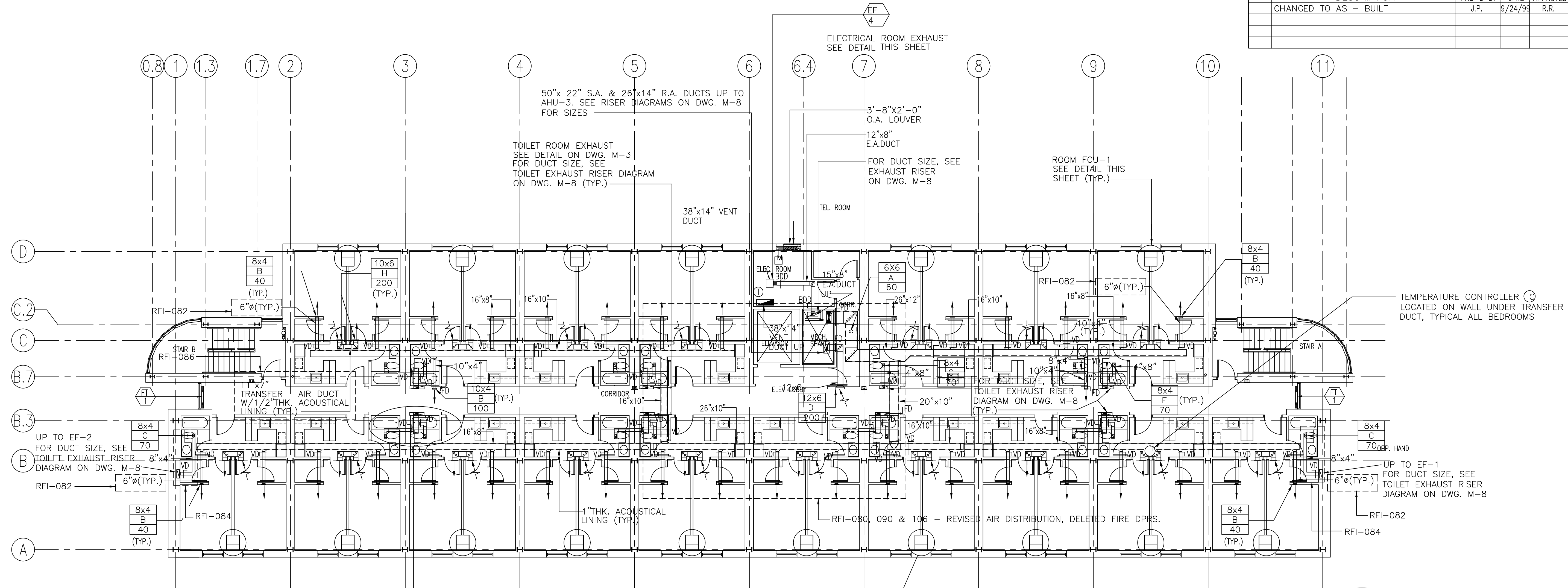
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F P-422 2191629 M-2

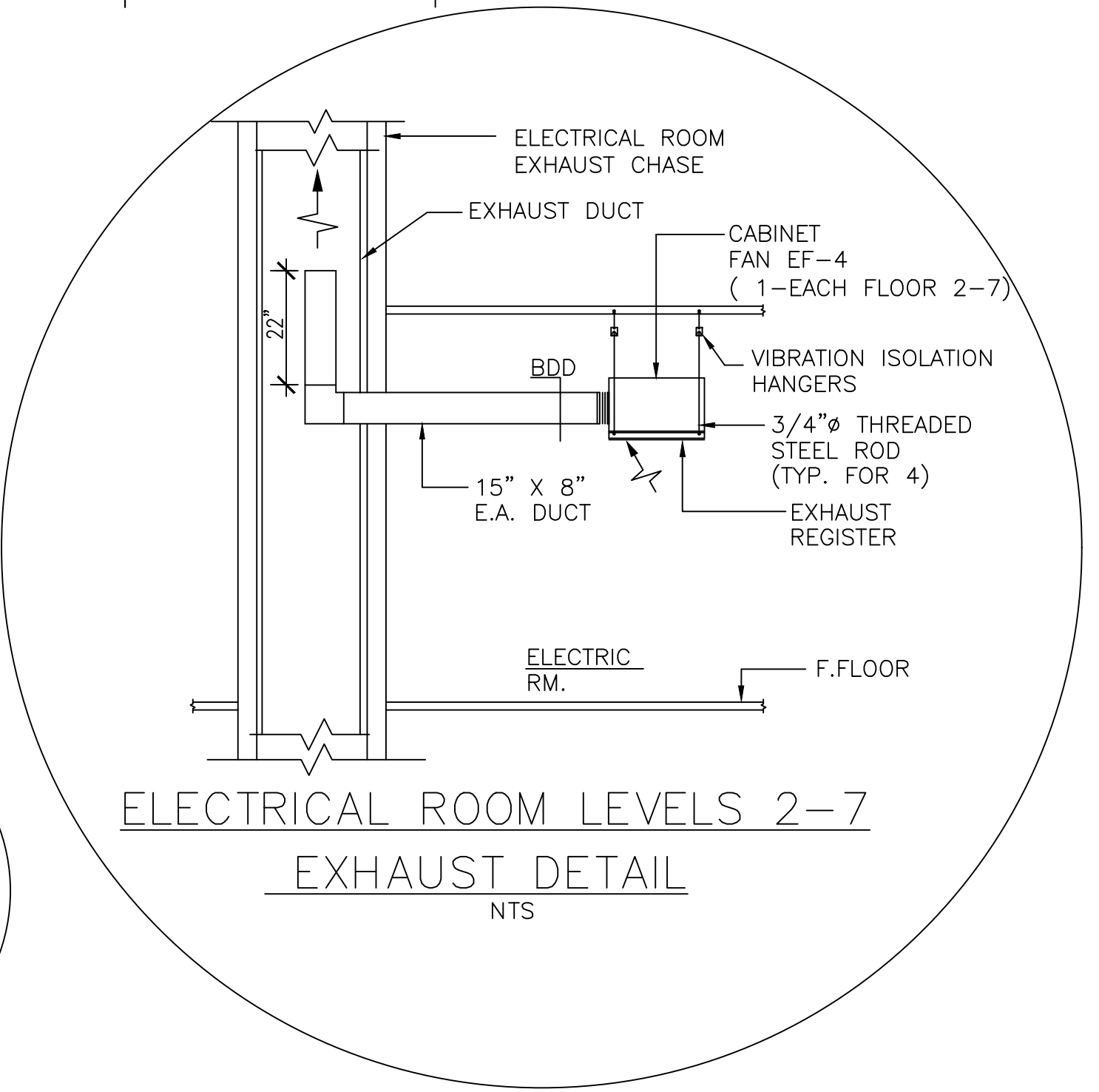
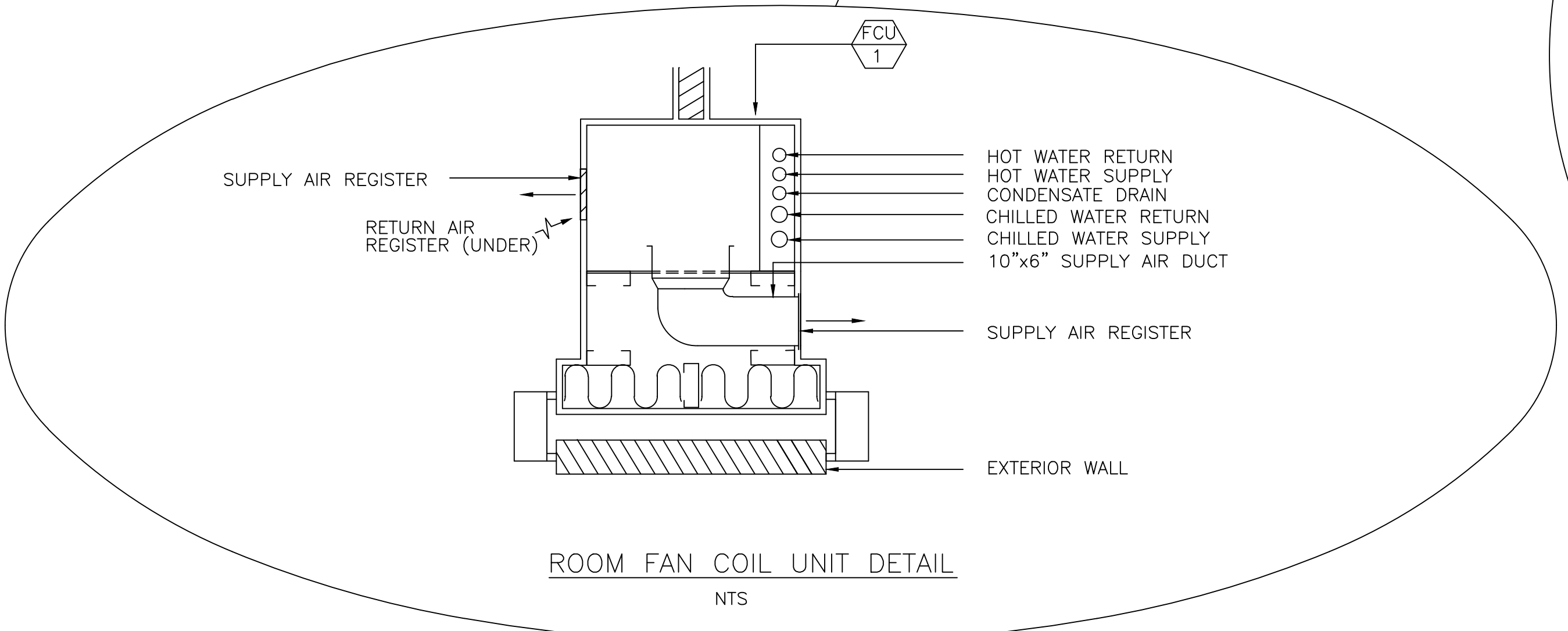
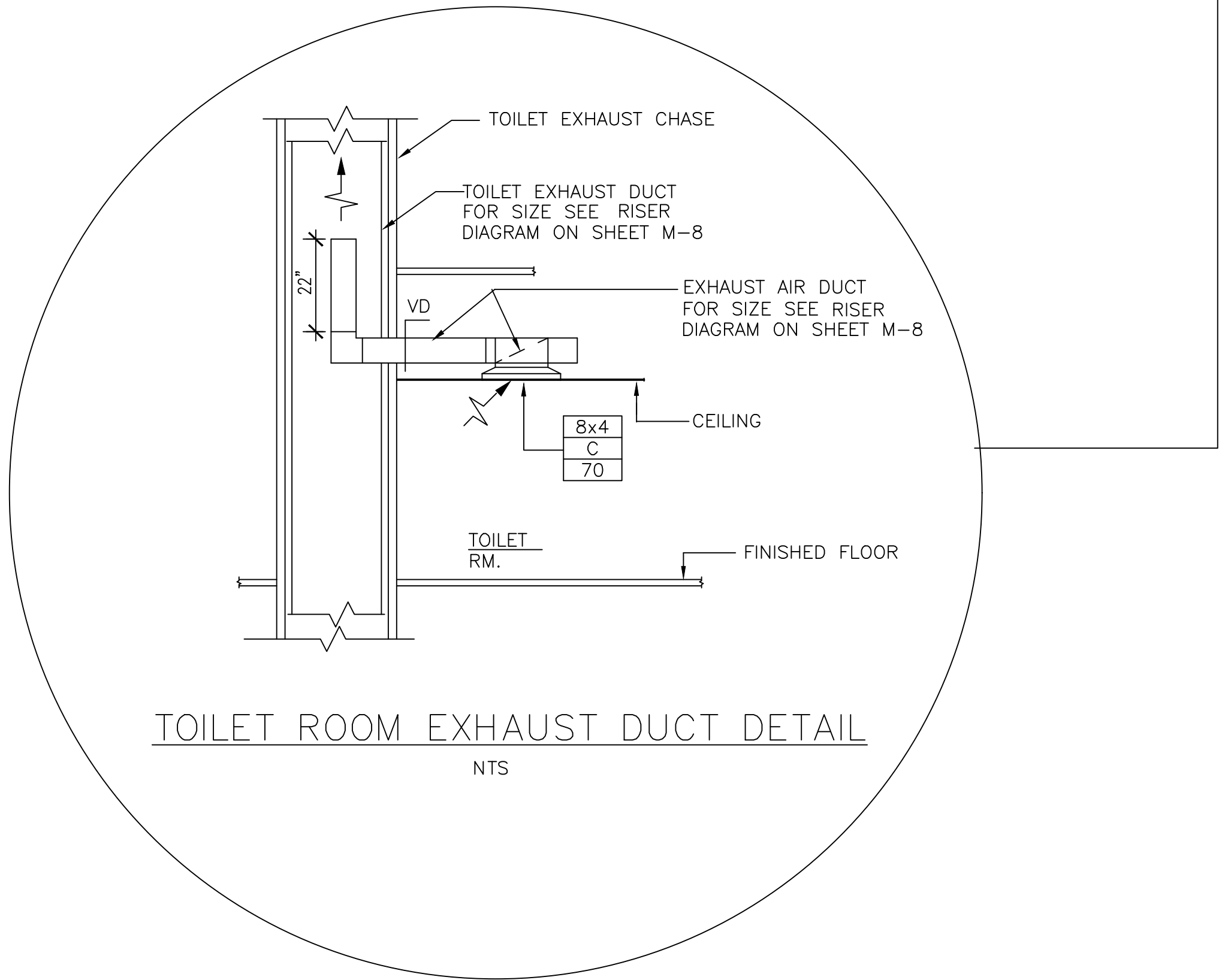
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SCALE AS NOTED SPEC. 04-96-0420 SHEET OF

REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	CHANGED TO AS - BUILT	J.P.	9/24/99	R.R.



LEVEL 2-7 PLAN
SCALE: 1/8"=1'-0"



graphic scales

CHECK GRAPHIC SCALES BEFORE USING

0 4' 8' 16'

SCALE: 1/8" = 1'

SATISFACTORY TO

DATE

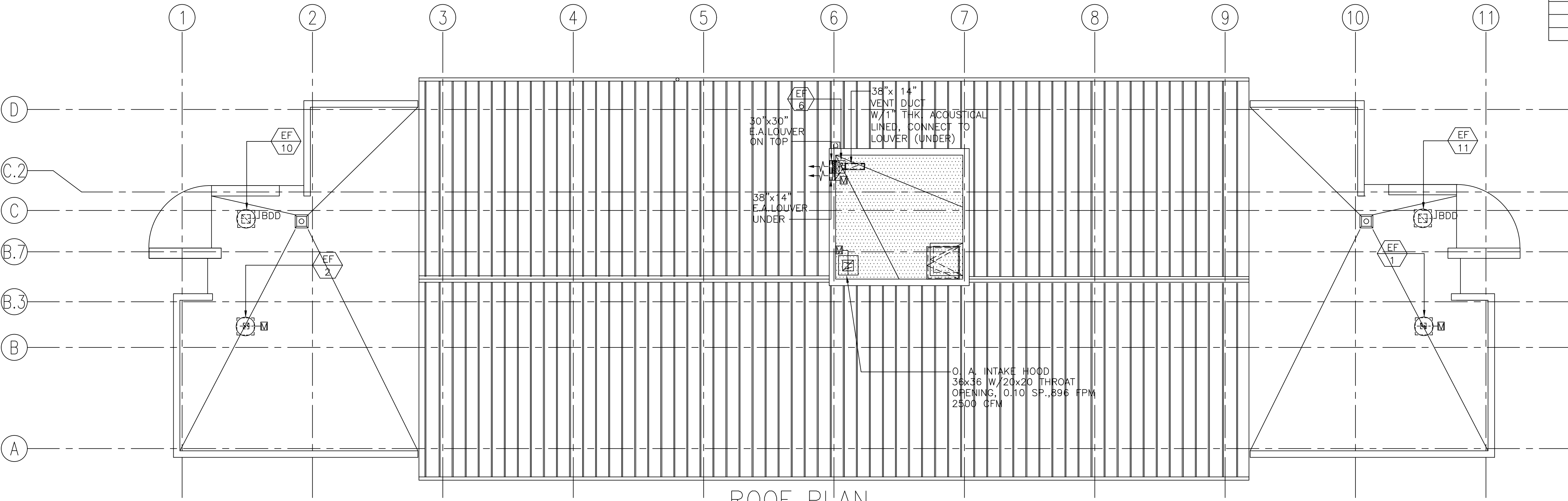
ARROWSTREET INC
SOMERVILLE, MA

DSGN R.D.L. DR R.D.L. CHK R.R.
SUPV R.R. CHK ENGR R.R.
SUBMITTED BY: W. DANIELL / PRINCIPAL
DATE: 28 MARCH 1997
FIRM MEMBER (TITLE)
NORTHDIV HD
DM HD
PFE DR
APPROVED DATE
OFFICER IN CHARGE DATE
NORTHDIV FOR COMMANDER NAVFAC

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
LESTER
NAVAL SUBMARINE BASE-NEW LONDON
PENNSYLVANIA
GROTON, CT
BACHELOR ENLISTED
QUARTERS
LEVEL 2 - 7 - DUCTWORK - PLAN

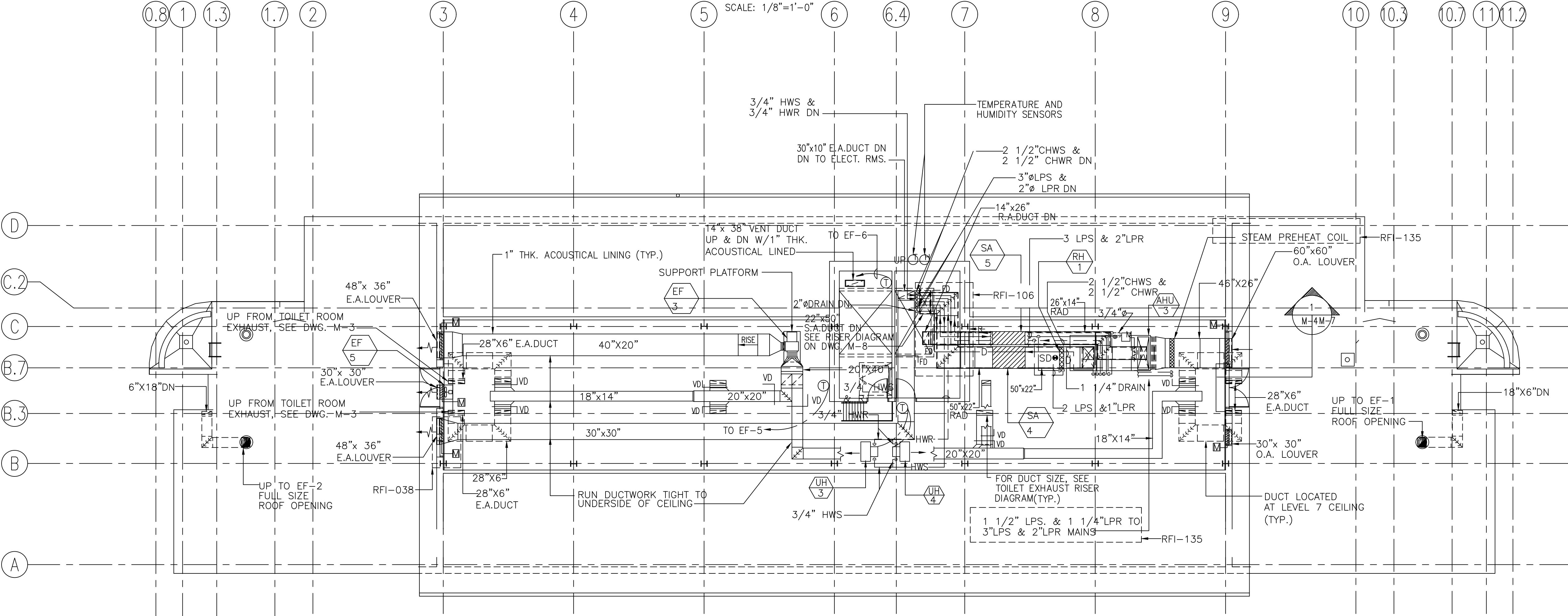
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CONSTR. CONTR. NO. N62472-96-C-0420
SPEC. 04-96-0420 SHEET 82 OF 135

REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	CHANGED TO AS - BUILT	J.P.	9/24/99	R.R.



ROOF PLAN

SCALE: 1/8"=1'-0"



LEVEL 8 - PLAN

SCALE: 1/8"=1'-0"

LEVEL 8 - PIPING & DUCTWORK -

graphic scales

CHECK GRAPHIC SCALES BEFORE USING

0 4' 8' 16'

SCALE: 1/8" = 1'

SATISFACTORY TO _____ DATE _____

ARROWSTREET INC
SOMERVILLE, MA

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
PENNSYLVANIA
NAVAL SUBMARINE BASE-NEW LONDON
GROTON, CT

LESTER

BACHELOR ENLISTED
QUARTERS

LEVEL 8 - PIPING & DUCTWORK - PLA

DSGN R.D.L. DR R.D.L. CHK R.R.
SUPV R.R. CHK ENGR R.R.

SUBMITTED BY: W. DANIELL / PRINCIPAL DATE 28 MARCH 1997

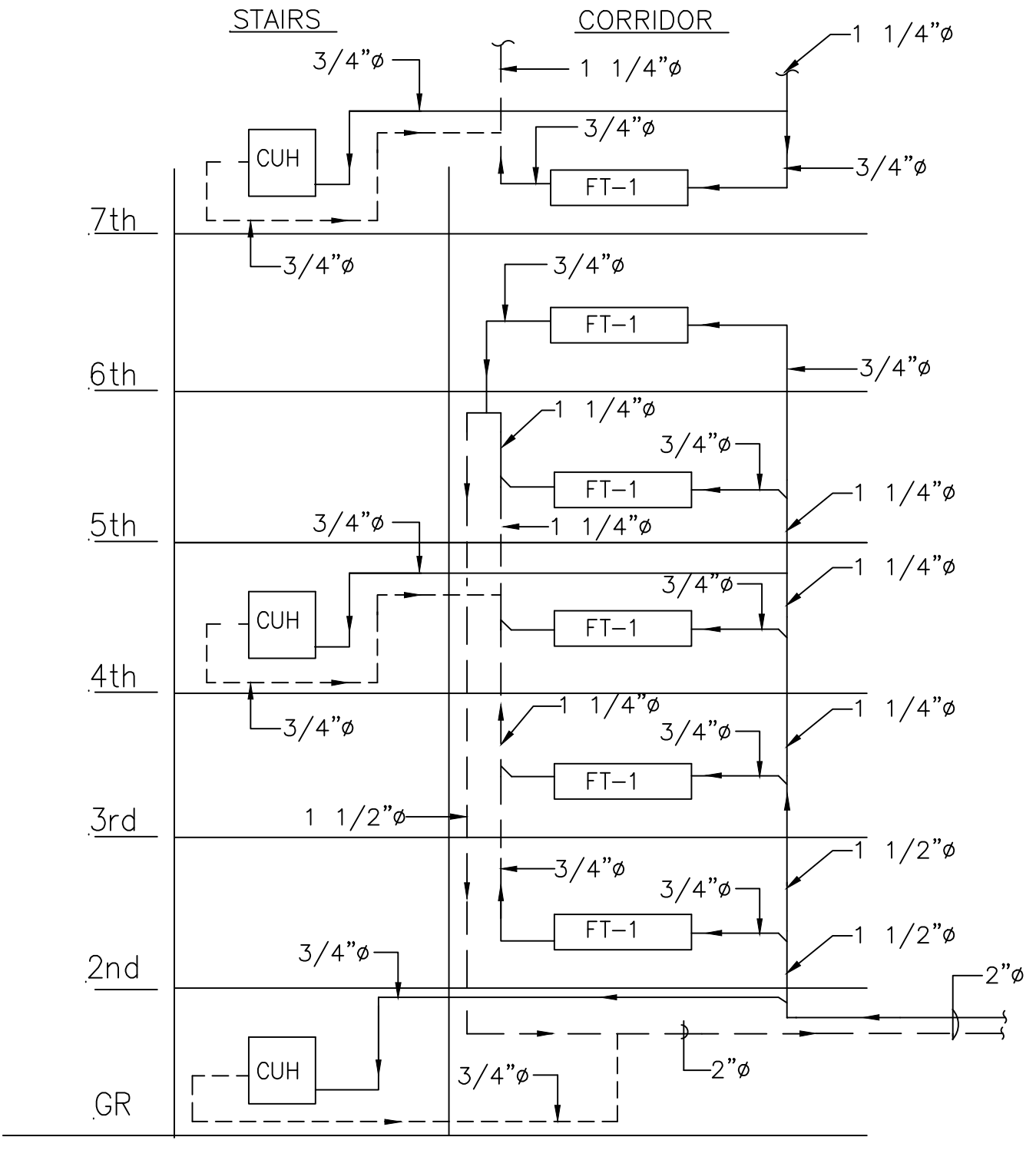
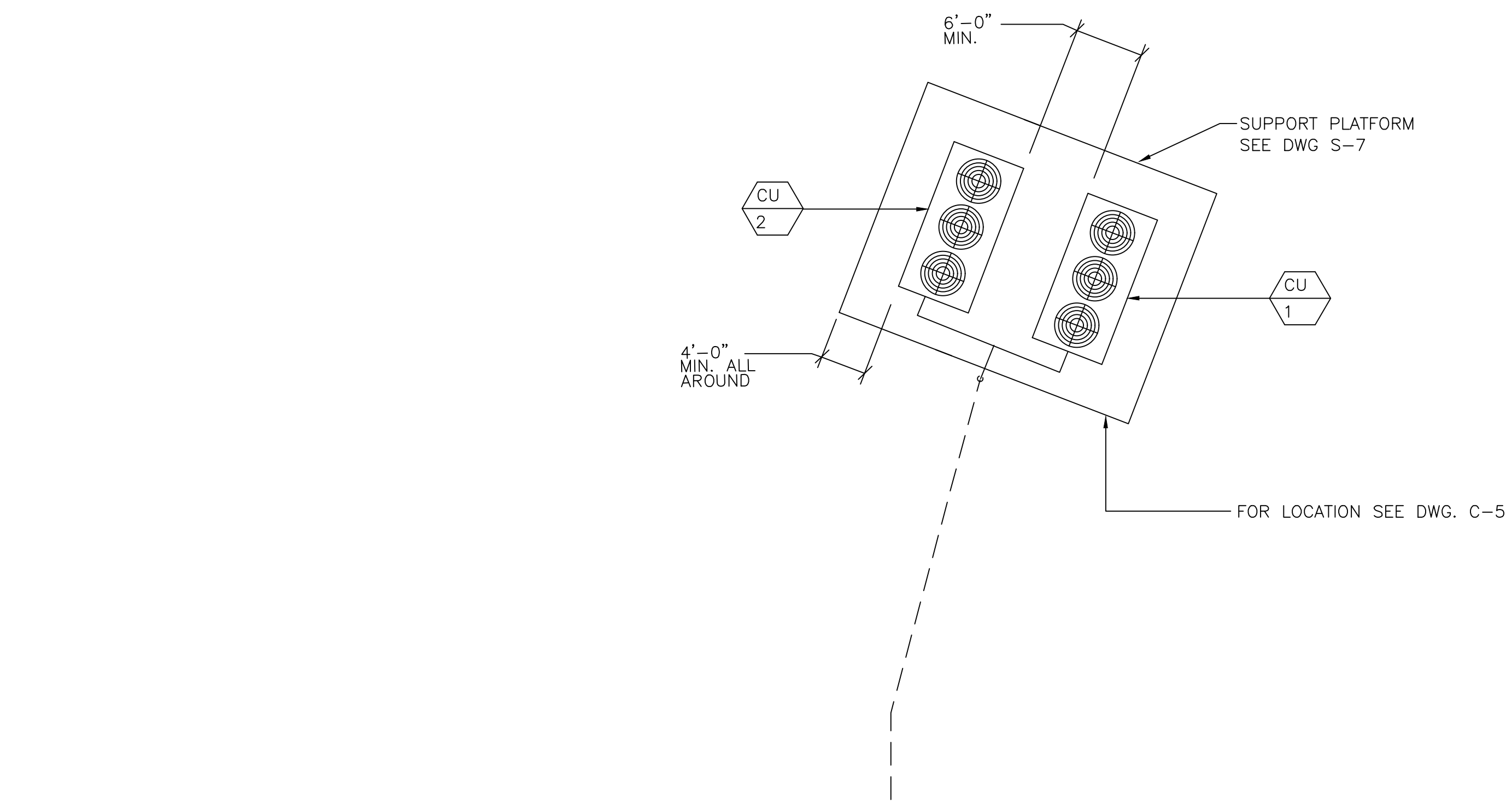
FIRM MEMBER (TITLE)
NORTHDIV HD
DM HD
PFE DR

APPROVED _____ DATE _____
OFFICER IN CHARGE

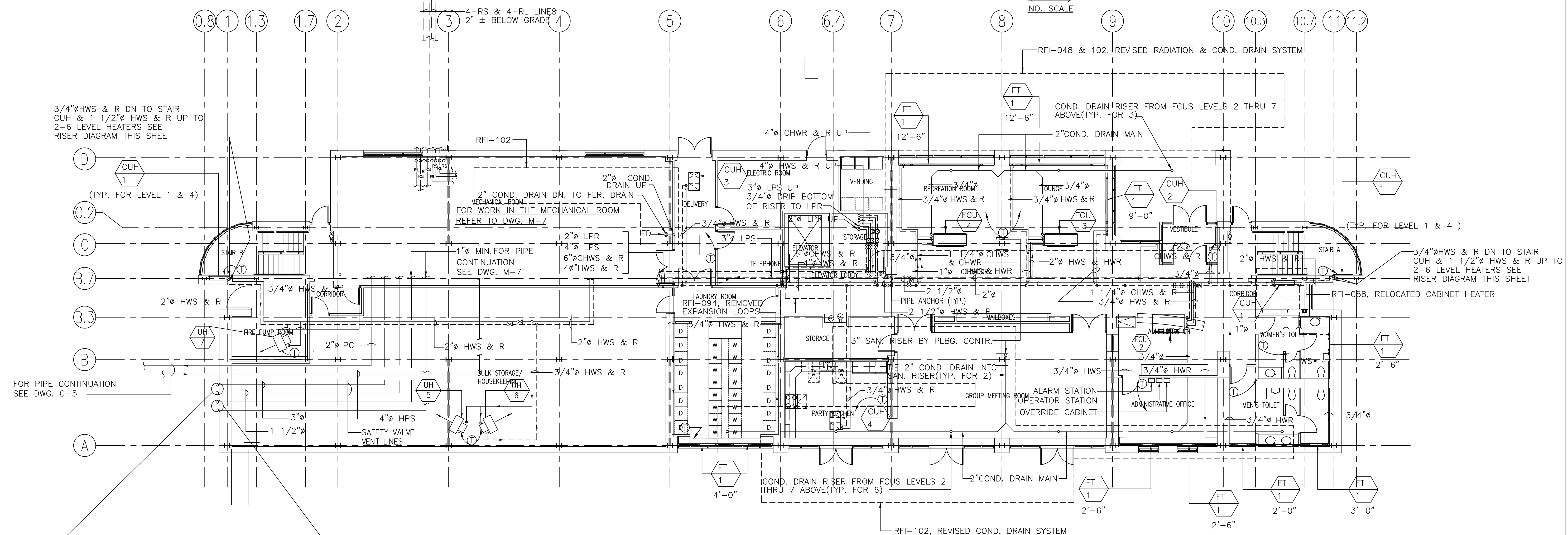
NORTHDIV FOR COMMANDER NAVFAC

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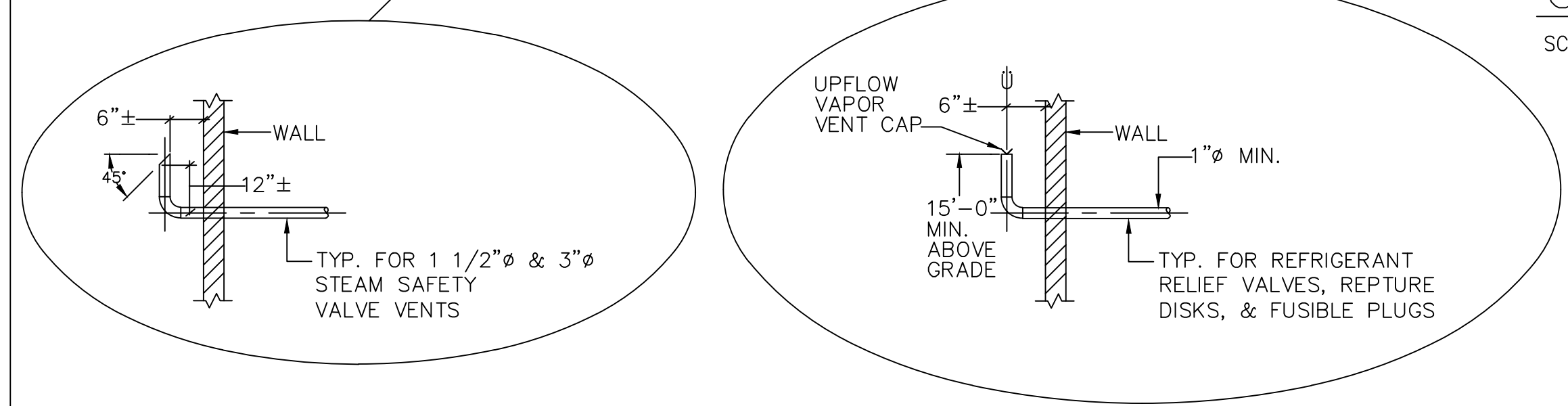
CONSTR. CONTR. NO. N62472-96-C-0420
SPEC. 04-96-0420 SHEET 83 OF 135



REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	CHANGED TO AS - BUILT	J.P.	9/24/99	R.R.



GROUND FLOOR-PIPING PLAN
SCALE: 1/8"=1'-0"



graphic scales

CHECK GRAPHIC SCALES BEFORE USING

0 4' 8' 16'

SCALE: 1/8" = 1'

SATISFACTORY TO _____ DATE _____

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NAVAL FACILITIES ENGINEERING COMMAND
NORTHERN DIVISION
LESTER
NAVAL SUBMARINE BASE-NEW LONDON
PENNSYLVANIA
GROTON, CT

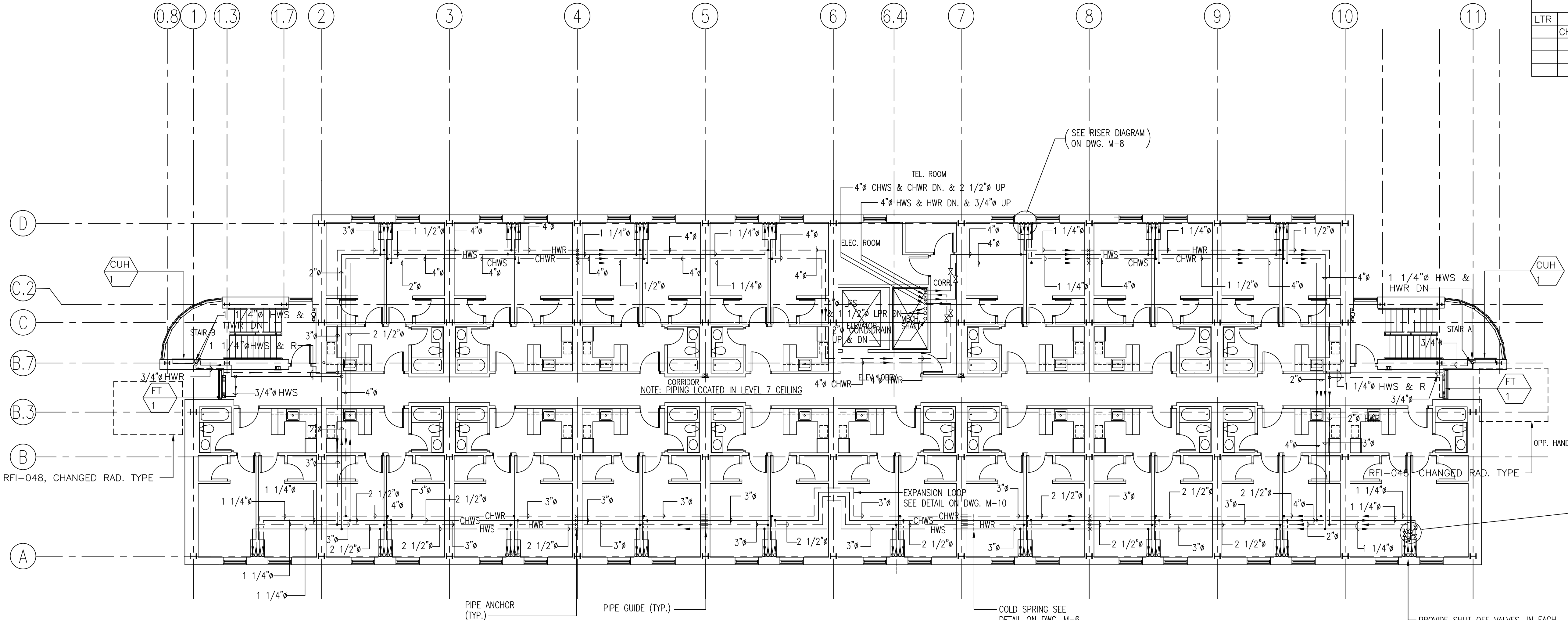
BACHELOR ENLISTED
QUARTERS
GROUND FLOOR - PIPING PLAN

SIZE F CODE IDENT. NO. P-422 NAVFAC DRAWING NO. 2191632 DIS. SH. NO. M-5

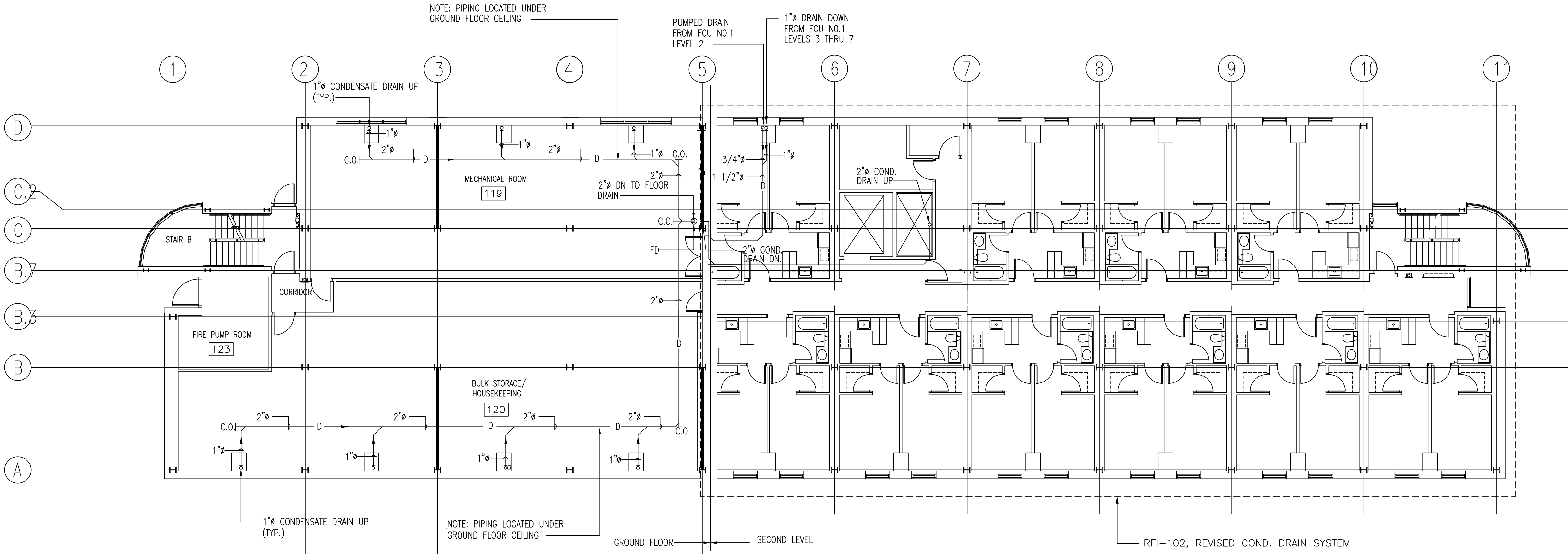
CONSTR. CONTR. NO. N62472-96-C-0420

SPEC. 04-96-0420 SHEET 84 OF 135

REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	CHANGED TO AS - BUILT	J.P.	9/24/99	R.R.



1 LEVEL 7 - PIPNG PLAN
M-6 M-6
SCALE: 1/8"=1'-0"



2 COOLING COIL CONDENSATE DRAIN PIPING PLAN
M-6 M-6
SCALE: 1/8"=1'-0"

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0 4' 8' 16'

SCALE: 1/8" = 1'

SATISFACTORY TO

DATE

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SOMERVILLE, MA

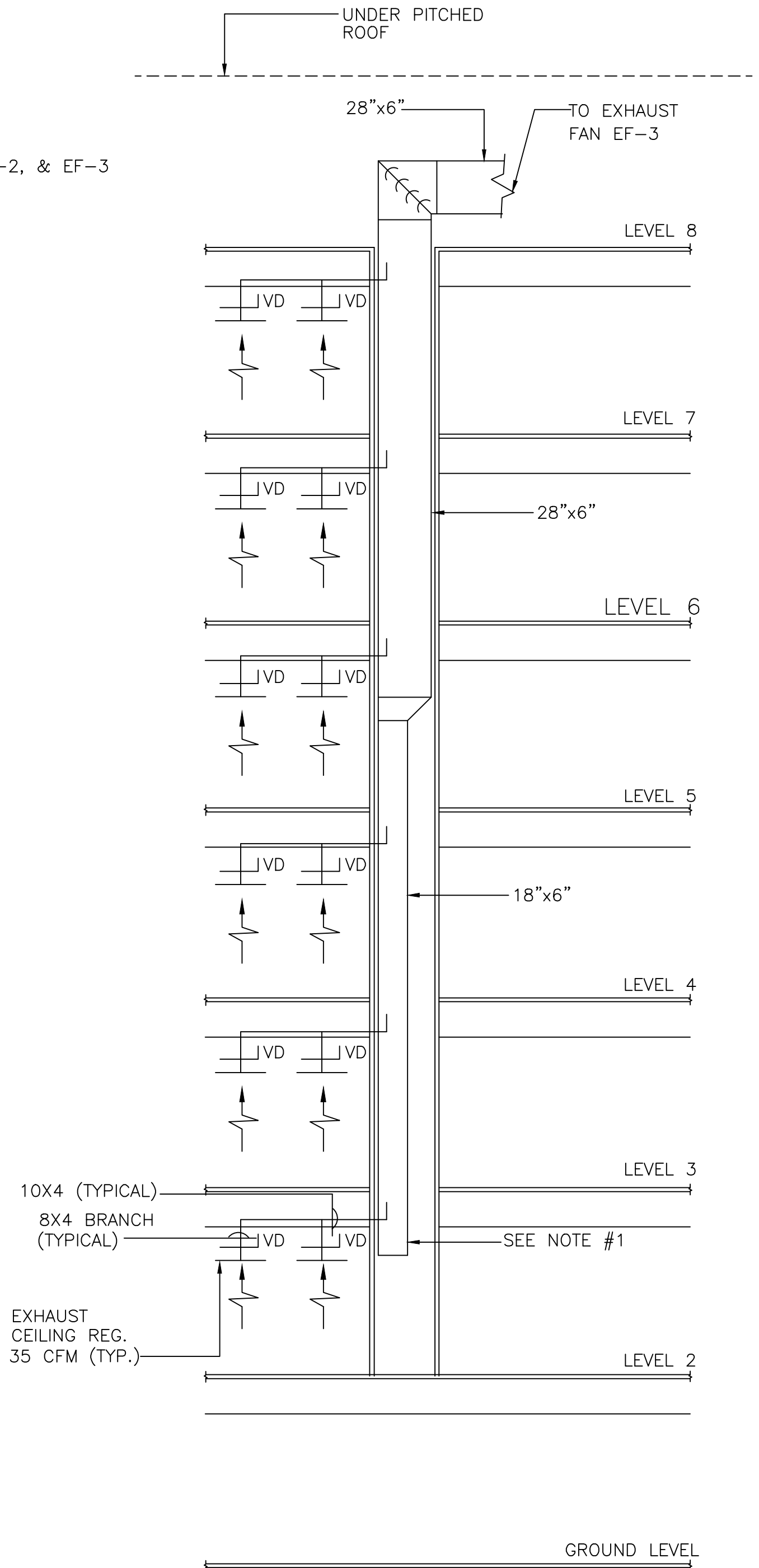
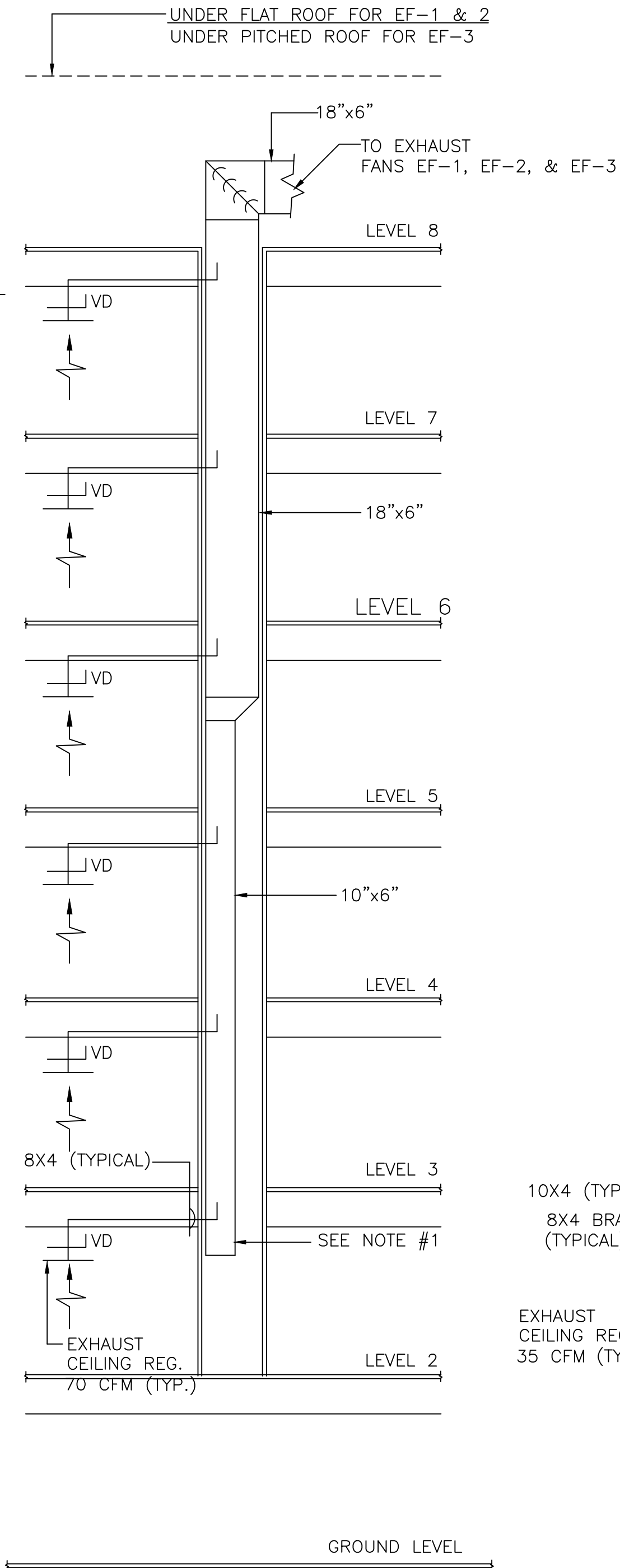
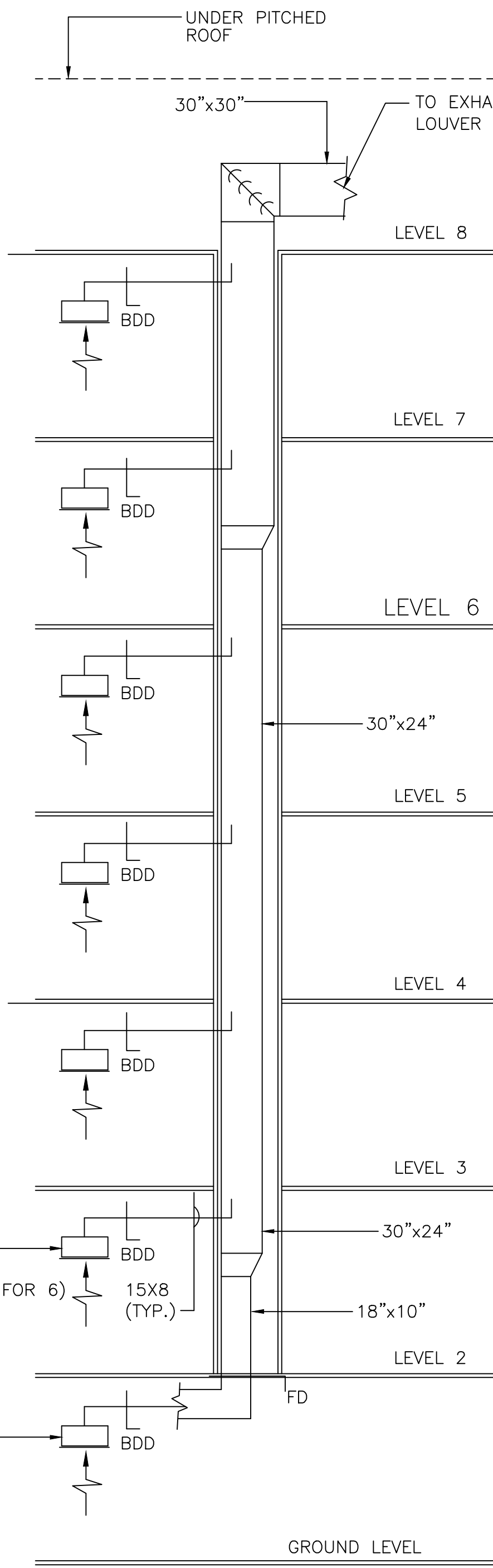
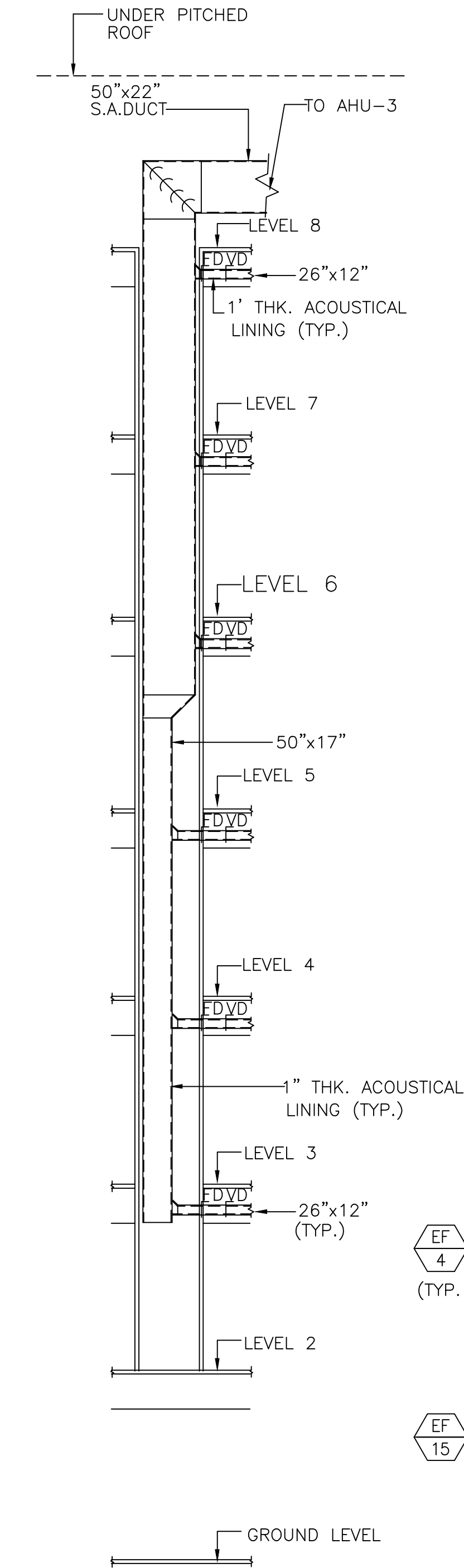
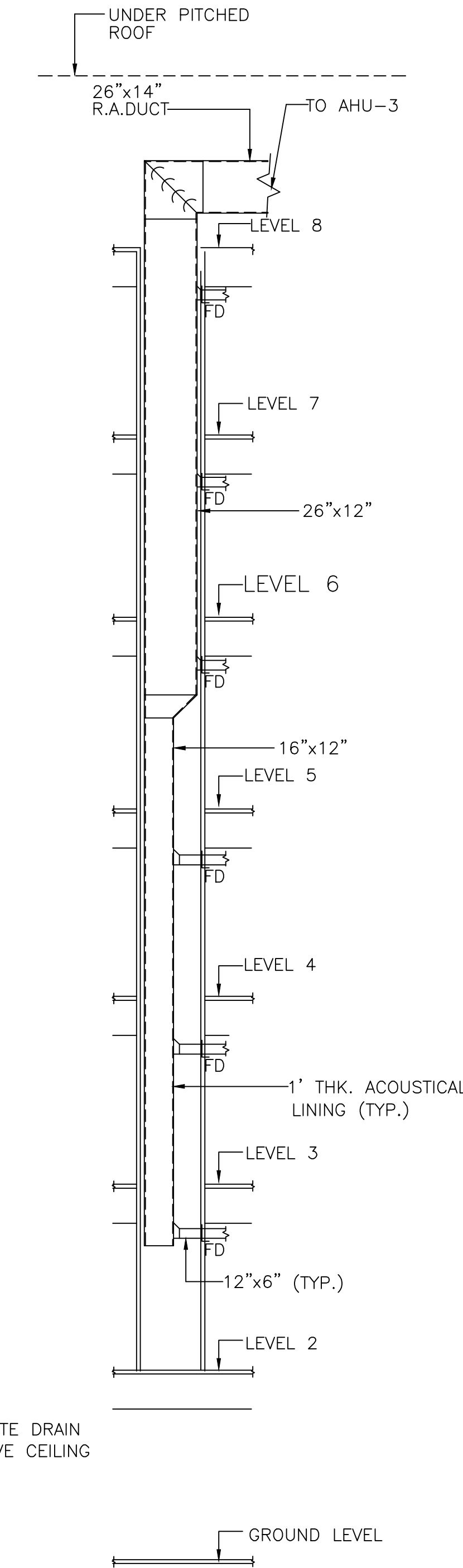
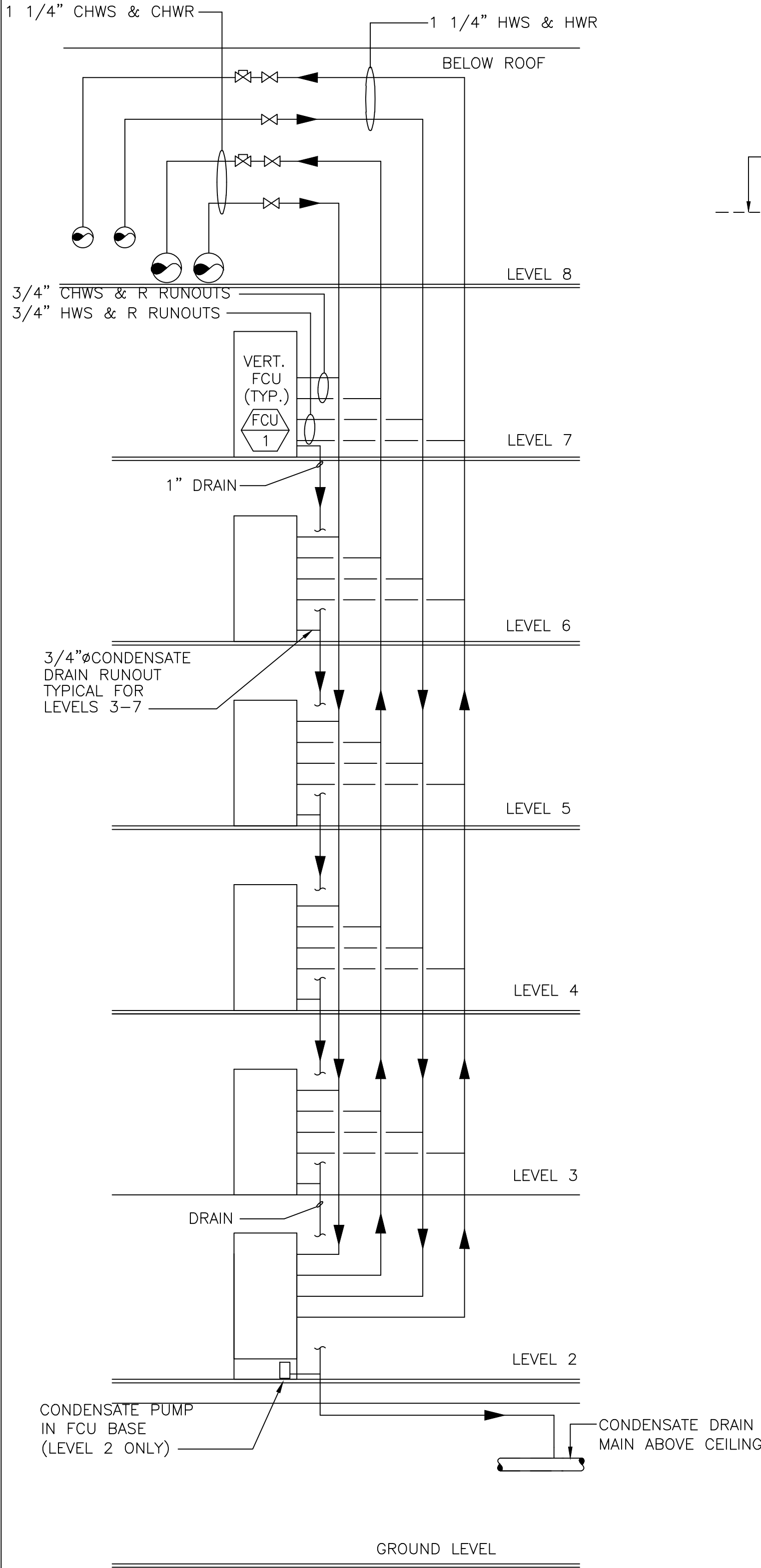
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL SUBMARINE BASE-NEW LONDON
GROTON, CT

DR R.D.L. DR R.D.L. CHK R.R.
SUPV R.R. CHK ENGR R.R.
SUBMITTED BY: W. DANIEL / PRINCIPAL DATE: 28 MARCH 1997
FIRM MEMBER (TITLE)
NORTHDIV HD
DM HD
PFE DRK
APPROVED DATE
OFFICER IN CHARGE DATE
APPROVED DATE
NORTHDIV FOR COMMANDER NAVFAC

NAVY FACILITIES ENGINEERING COMMAND
NAVAL SUBMARINE BASE-NEW LONDON
GROTON, CT
BACHELOR ENLISTED
QUARTERS
LEVEL 2 & 7 - PIPING PLAN

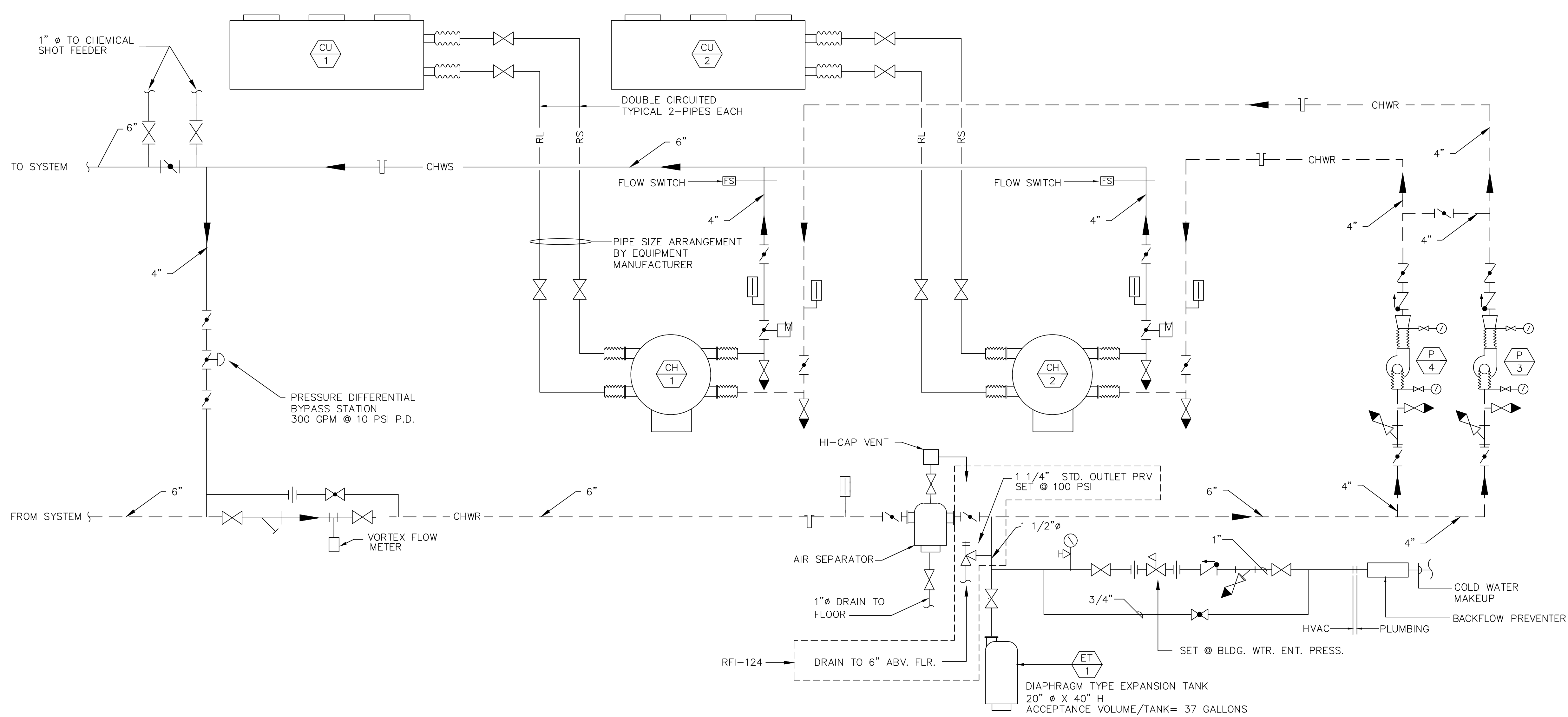
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NAVFAC DRAWING NO.: 2191633
CONSTR. CONTR. NO. N62472-96-C-0420
SHEET 85 OF 135

REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	AS-BUILT, NO CHANGES	J.P.	9/24/99	R.R.



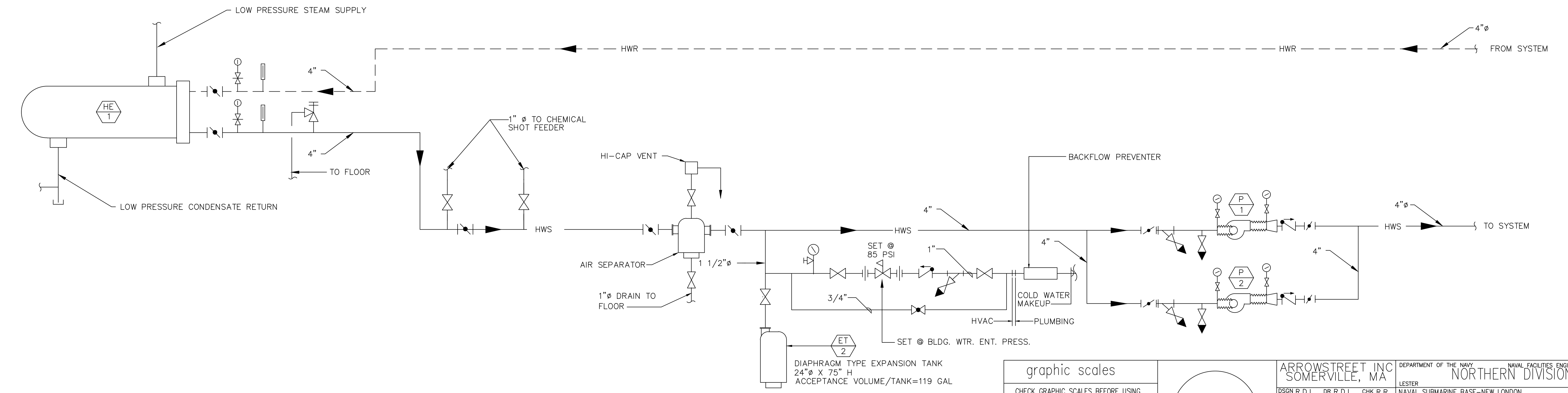
NOTE:
1. COAT INSIDE BOTTOM 12" SECTION OF EXHAUST DUCT RISERS WITH 2-COATS OF A EPOXY-POLYAMIDE PAINT.

graphic scales			ARROWSTREET INC SOMERVILLE, MA		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND	
CHECK GRAPHIC SCALES BEFORE USING			NORTHERN DIVISION		NAVAL SUBMARINE BASE-NEW LONDON PENNSYLVANIA GROTON, CT	
			DSGN R.D.L. DR R.D.L. CHK R.R. SUPY R.R. CHK ENGR R.R.		NAVAL FACILITIES ENGINEERING COMMAND	
			SUBMITTED BY: W. DANIEL / PRINCIPAL DATE: 28 MARCH 1997		NAVAL SUBMARINE BASE-NEW LONDON	
			FIRM MEMBER (TITLE)		NAVAL FACILITIES ENGINEERING COMMAND	
			NORTH DIV HD		NAVAL FACILITIES ENGINEERING COMMAND	
			DM HD		NAVAL FACILITIES ENGINEERING COMMAND	
			PPE DR		NAVAL FACILITIES ENGINEERING COMMAND	
			APPROVED DATE		NAVAL FACILITIES ENGINEERING COMMAND	
			OFFICER IN CHARGE DATE		NAVAL FACILITIES ENGINEERING COMMAND	
			APPROVED DATE		NAVAL FACILITIES ENGINEERING COMMAND	
			NORTH DIV FOR COMMANDER NAVFAC		NAVAL FACILITIES ENGINEERING COMMAND	
			SIZE F		NAVAL FACILITIES ENGINEERING COMMAND	
			CODE IDENT. NO. P-422		NAVAL FACILITIES ENGINEERING COMMAND	
			NAVIFAC DRAWING NO. 2191635		NAVAL FACILITIES ENGINEERING COMMAND	
			DIS. SH. NO. M-8		NAVAL FACILITIES ENGINEERING COMMAND	
			CONSTR. CONTR. NO. N62472-96-C-0420		NAVAL FACILITIES ENGINEERING COMMAND	
			SPEC. 04-96-0420		NAVAL FACILITIES ENGINEERING COMMAND	
			SHEET 87 OF 135		NAVAL FACILITIES ENGINEERING COMMAND	



CHILLED WATER PIPING FLOW DIAGRAM

SCALE: NONE



HOT WATER PIPING FLOW DIAGRAM

SCALE: NONE

REVISIONS				
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SATISFACTORY TO

DATE

ARROWSTREET INC
SOMERVILLE, MA

DSGN R.D.L. DR R.D.L. CHK R.R.
SUPV R.R. CHK ENGR R.R.
SUBMITTED BY: W. DANIEL / PRINCIPAL DATE: 28 MARCH 1997
FIRM MEMBER (TITLE)
NORTHDIV _____ HD _____
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NORTHDIV FOR COMMANDER NAVFAC

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
LESTER
NAVAL SUBMARINE BASE-NEW LONDON
PENNSYLVANIA
GROTON, CT

NORTHERN DIVISION
BACHELOR ENLISTED
QUARTERS
FLOW DIAGRAMS

SIZE

CODE IDENT. NO.

NAVFAC DRAWING NO.

DIS. SH. NO.

F

P-422

2191636

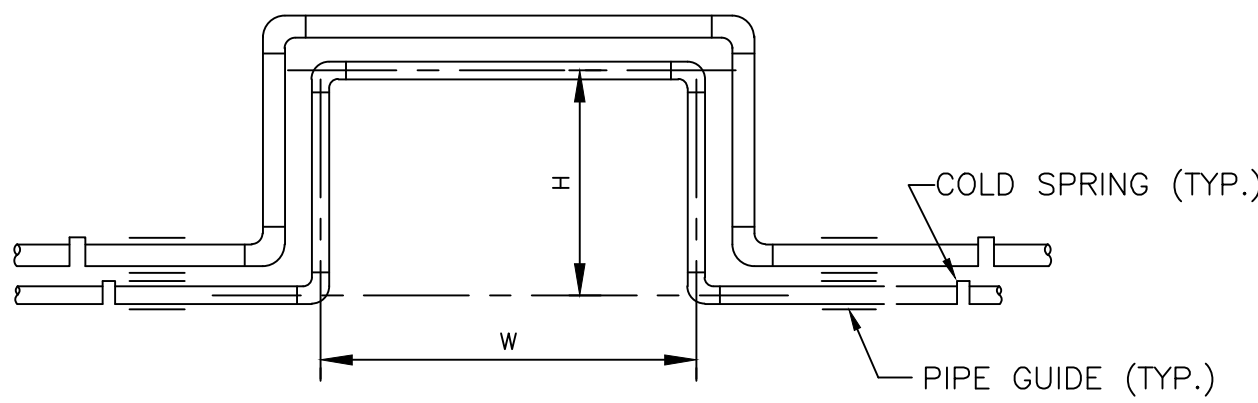
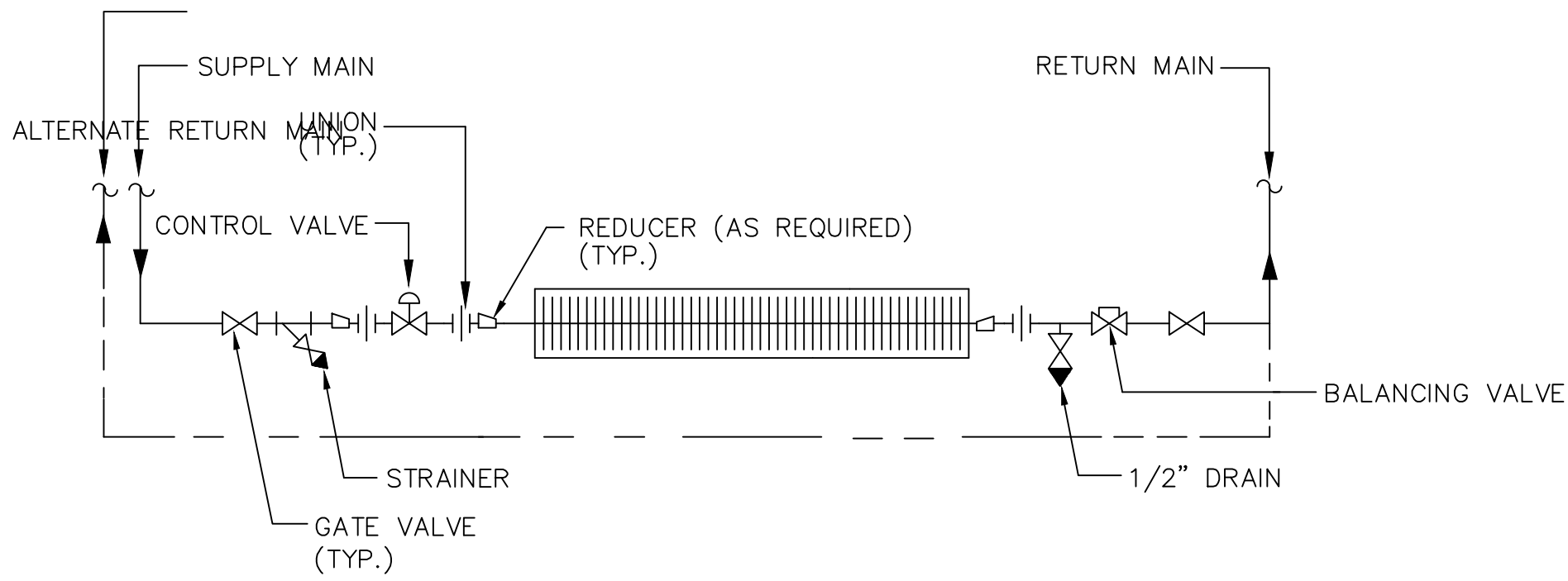
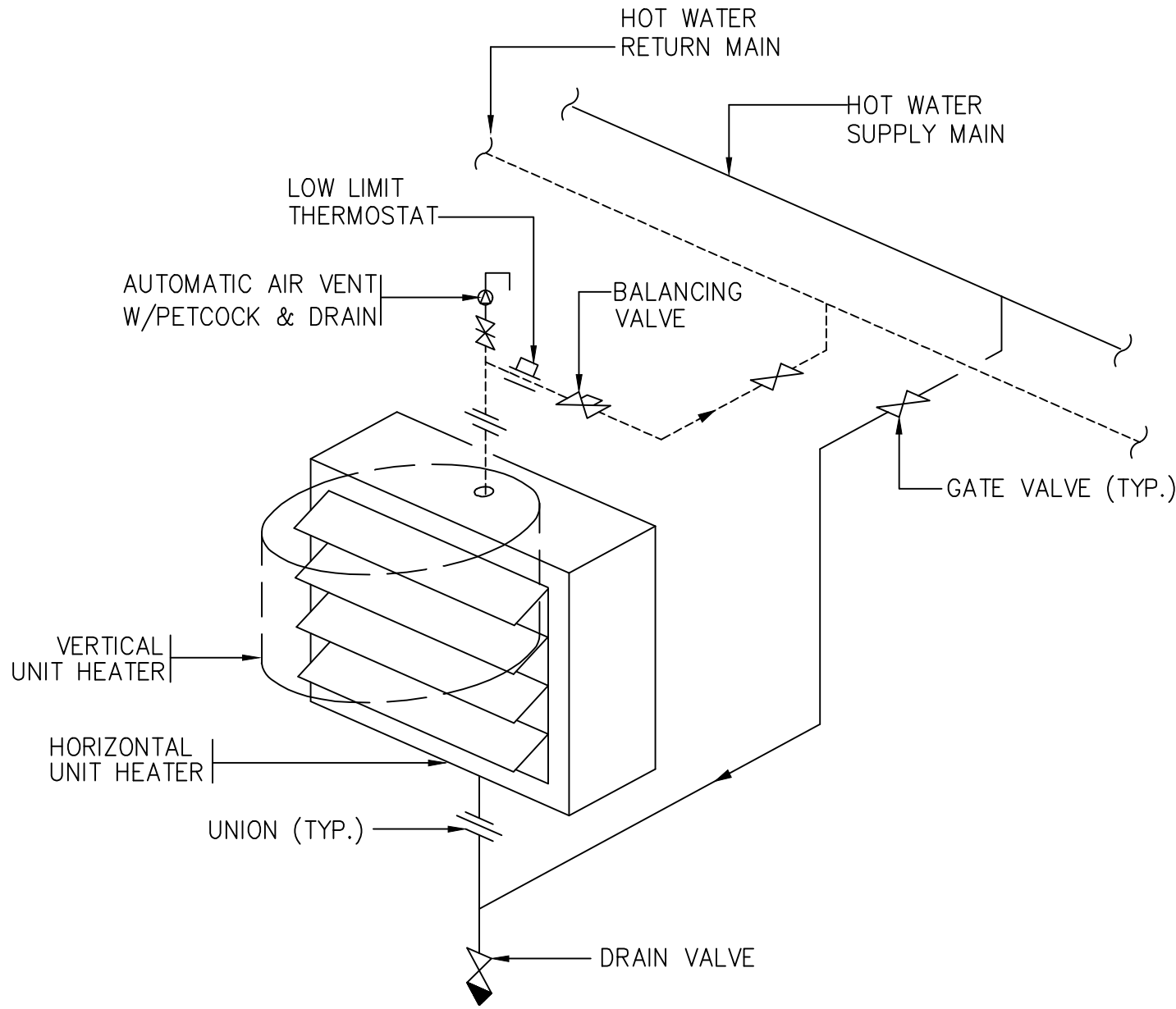
M-9

CONSTR. CONTR. NO. N62472-96-C-0420

SPEC. 04-96-0420

SHEET 88 OF 135

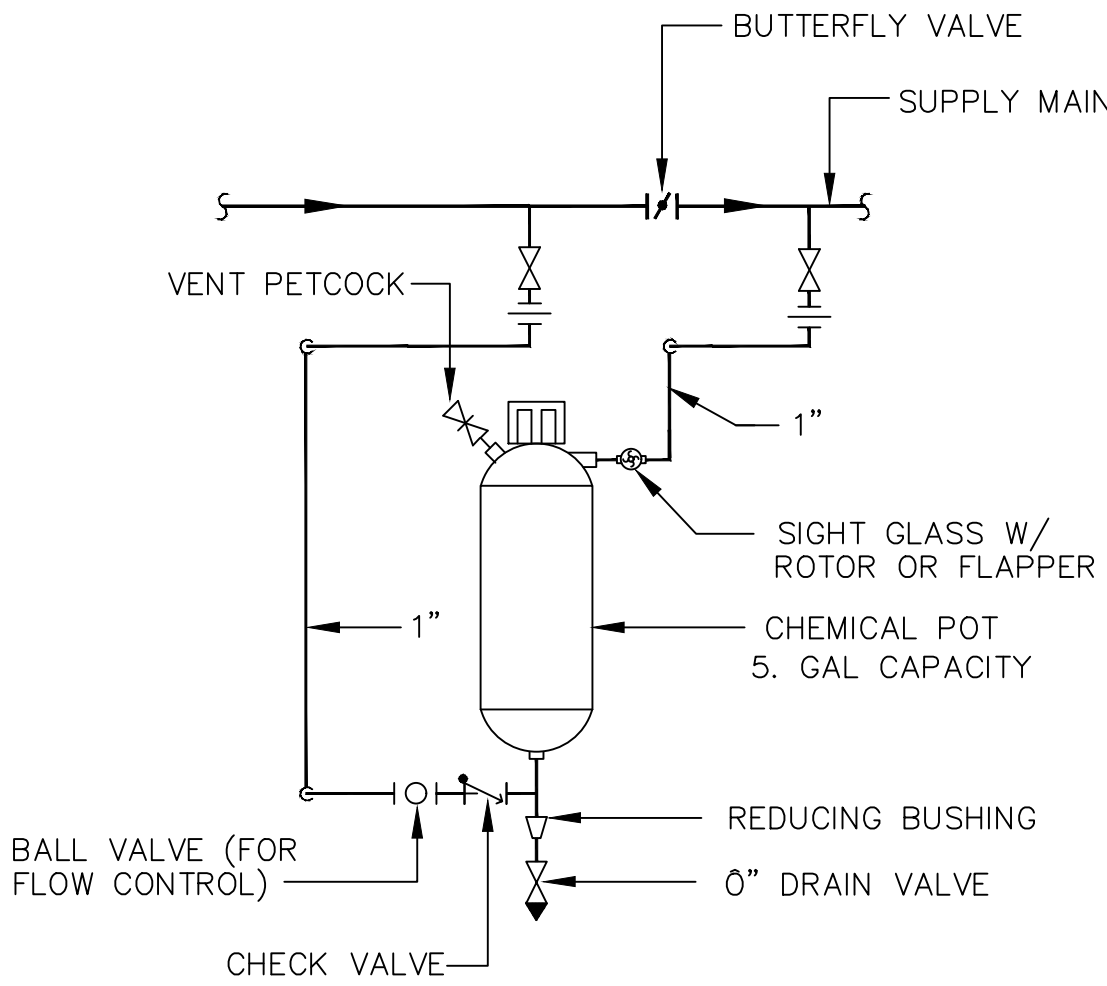
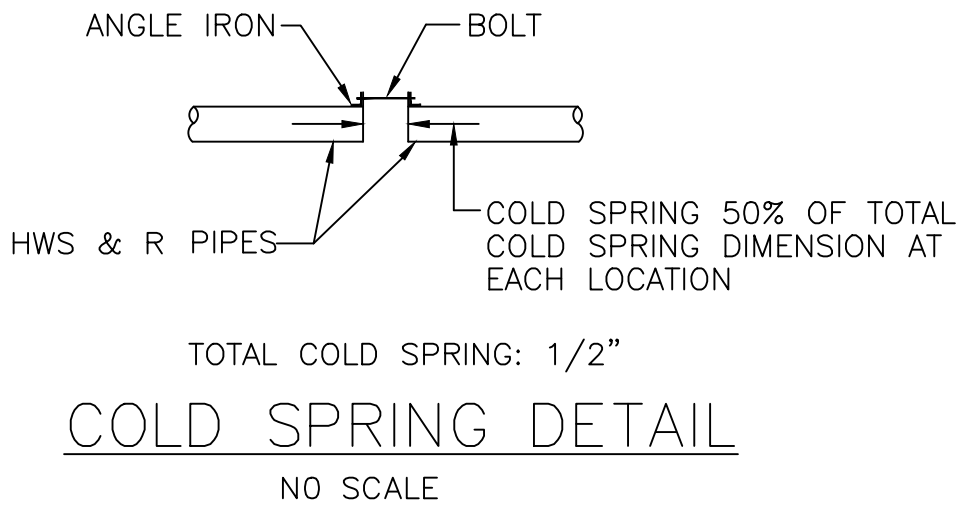
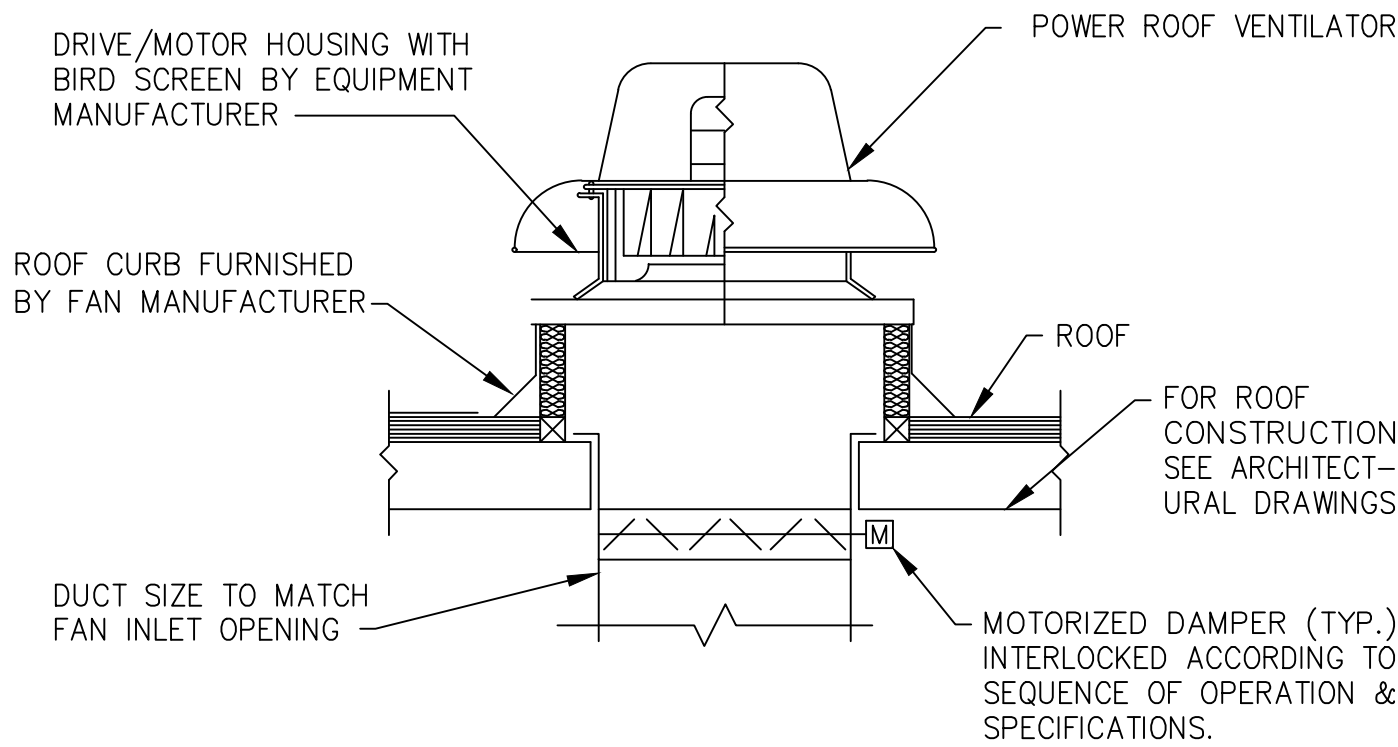
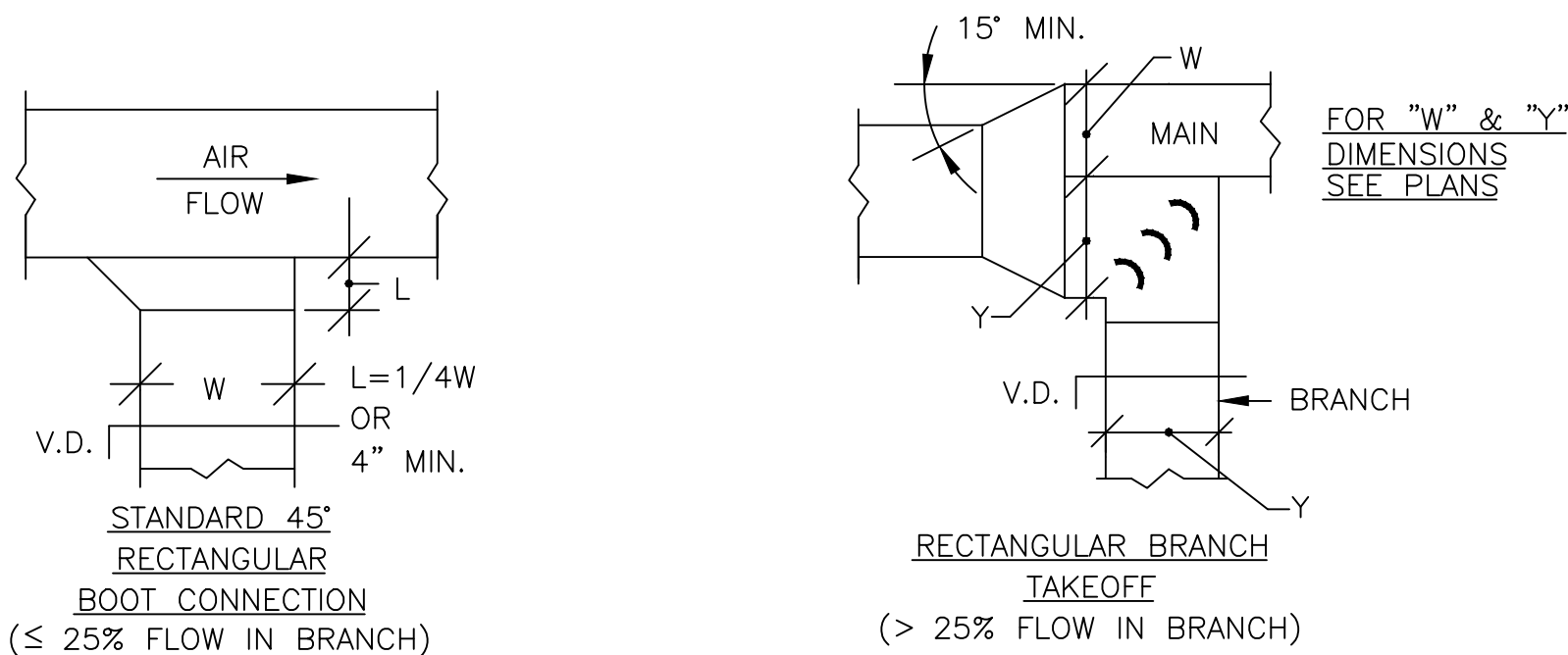
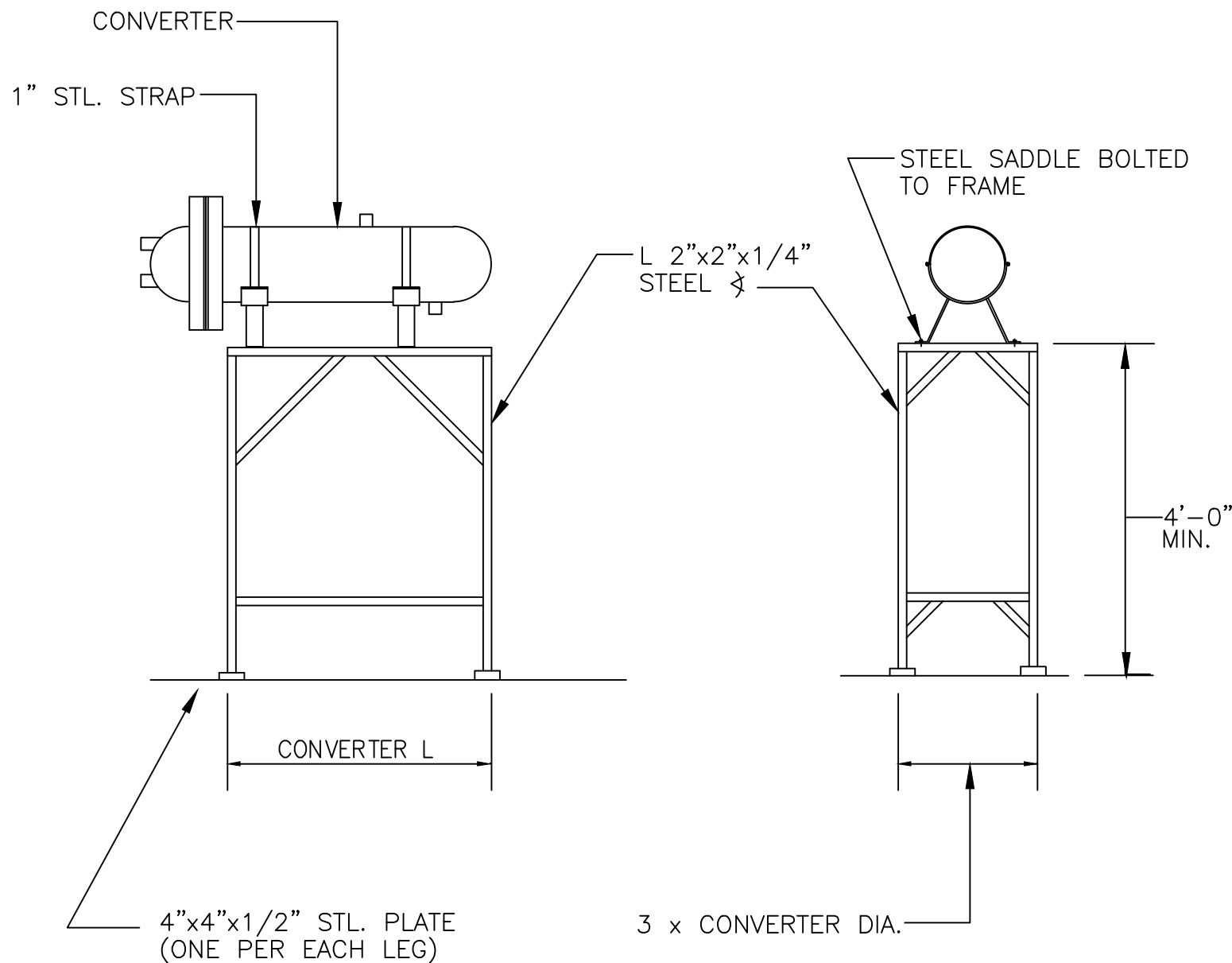
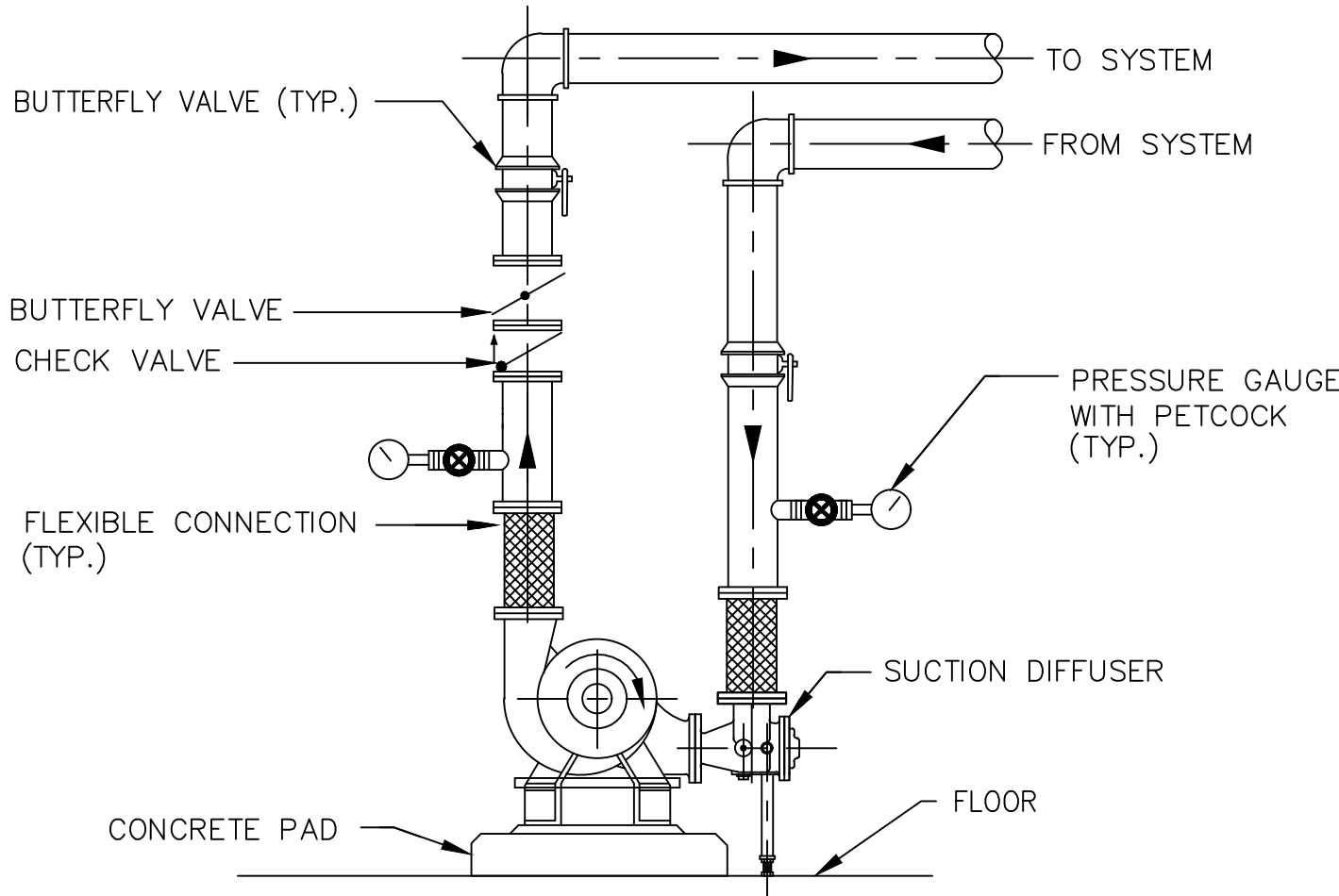
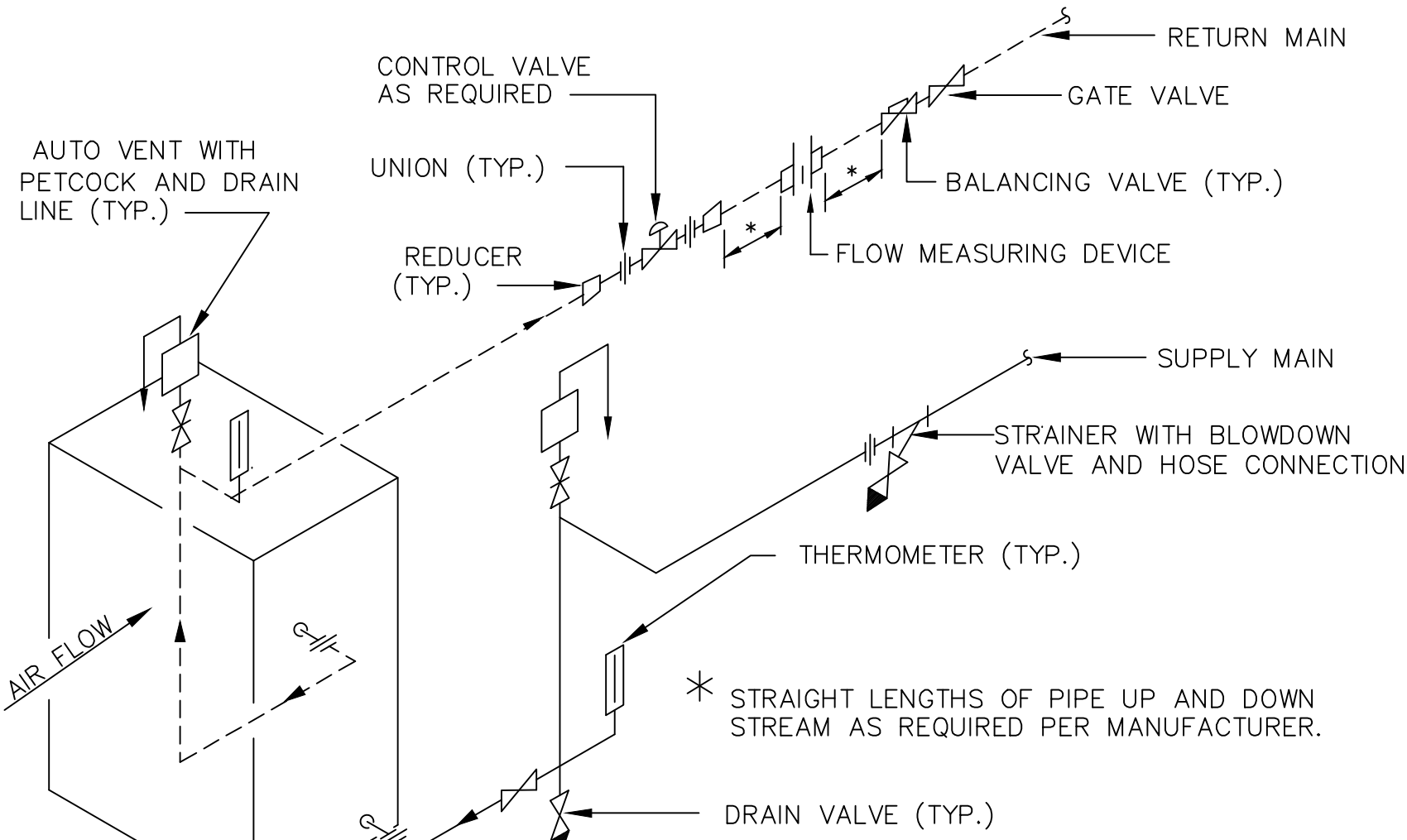
REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	AS-BUILT, NO CHANGES	J.P.	9/24/99	R.R.



PIPE SIZE	W	H
UP TO 6"	6 FEET	3 FEET
6"	8 FEET	4 FEET

NOTE: DIMENSIONS SHOWN IS FOR SMALLEST LOOP

HOT WATER UNIT HEATER PIPING DETAIL
NO SCALE



graphic scales		ARROWSTREET INC SOMERVILLE, MA		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND	
CHECK GRAPHIC SCALES BEFORE USING		NORTHERN DIVISION		LESTER	
		BACHELOR ENLISTED QUARTERS DETAILS		NAVAL SUBMARINE BASE-NEW LONDON PENNSYLVANIA GROTON, CT	
		DSGN R.D.L. DR R.D.L. CHK R.R. SUPV R.R. CHK ENGR R.R.		NAVFAC DRAWING NO. 2191637	
		SUBMITTED BY: W. DANIELL / PRINCIPAL DATE: 28 MARCH 1997		DIS. SH. NO. M-10	
		FIRM MEMBER (TITLE)		CONSTR. CONTR. NO. N62472-96-C-0420	
		NORTHDIV DM HD PPE		SPEC. 04-96-0420 SHEET 89 OF 135	
		APPROVED DATE		SCALE NONE	
		OFFICER IN CHARGE		SIZE F P-422	
		APPROVED DATE			
		NORTHDIV FOR COMMANDER NAVFAC			

REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	AS-BUILT, NO CHANGES	J.P.	9/24/99	R.R.

AIR HANDLING UNIT SCHEDULE																													
SYMBOL	SERVICE	FAN DATA				COOLING COIL DATA												HEATING COIL DATA								FILTERS			REMARKS
		CFM	EXT. S.P.	HP	VOLT/PH	EDB (°F)	EWB (°F)	LDB (°F)	LWB (°F)	MAX FV (FPM)	MAX (APD)	TOTAL MBH	SENS. MBH	FLUID	EWT (°F)	LWT (°F)	GPM	MAX (WPD)	AIR			STEAM			MAX FV (FPM)	APD	TYPE		
																			EAT (°F)	LAT (°F)	MAX (APD)	PSIG	LBS./HR.	MAX (SPD)				TOTAL MBH	
AHU-1	MEETING ROOM	2400	0.88	5	460/3	81.3	69.6	54.6	54.3	500	0.53	117.2	70.8	WATER	45	54.3	23.4	5.23	28.7	76.8	0.10	2	129.4	0.24	125.1	400	0.5"	2"-30%	SEE NOTE
AHU-2	LAUNDRY ROOM	6848	0.75	5	460/3	—	—	—	—	—	—	—	—	—	—	—	—	—	5	68	0.20	2	483.1	0.73	467.9	400	0.5"	2"-30%	
AHU-3	LIVING/QUARTERS	9400	3.0	10	460/3	83.7	70.8	60.0	58.8	500	0.41	381.4	246.7	WATER	46	56	76.3	13.4	13	67.7	0.16	2	576.5	0.55	558	400	0.5"	2"-30%	
RH-1	AHU-3 REHEAT	9400	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	60	75	0.25	2	159.0	—	152.3	—	—	—	INSTALL IN DUCTWORK

NOTE :
FULL & 2/3 SPEED
FACE & BYPASS DAMPER

FAN COIL UNIT																				
SYMBOL	SERVICE	TYPE	NOMINAL CAPACITY (C.F.M.)	EXT. S.P.	FAN MOTOR (H.P.)	COOLING								HEATING						REMARKS
						SENSIBLE CAPACITY (MBH)	TOTAL CAPACITY (MBH)	AIR TEMP.(°F)				ENTERING WATER		COIL PRESS. DROP (FT. H ₂ O)	CAPACITY (B.T.U.H.)	TEMP.(°F)		WATER RATE (G.P.M.)	COIL PRESS. DROP (FT. H O)	
								EDB	EWB	LDB	LWB	TEMP. (°F)	RATE (G.P.M.)			EAT	EWT			
FCU-1	LIVING/QUARTERS	FLOOR MTD.	360	0.10	1/10	7.9	10.6	75.0	62.0	54.6	51.4	45.0	1.5	4.1	10,700	68	160	1.0	1.8	SEE NOTE
FCU-2	RECEPTION/ADMIN.	HORIZ. CEILING MTD.	1360	0.50	1/4	31.1	40.2	75.2	63.2	54.0	52.9	45.0	9.8	5.0	19,700	60	160	1.0	1.6	
FCU-3	VENDING/REC.	HORIZ. CEILING MTD.	1360	0.50	1/4	31.1	40.2	75.2	63.2	54.0	52.9	45.0	9.8	5.0	13,800	60	160	1.0	1.6	
FCU-4	LOUNGE	HORIZ. CEILING MTD.	1360	0.50	1/4	31.1	40.2	75.2	63.2	54.0	52.9	45.0	9.8	5.0	13,800	60	160	1.0	1.6	

NOTE :
600 RPM UNIT
OPERATING AT LOW SPEED

REGISTERS, GRILLES & DIFFUSERS						
SYMBOL	DEVICE	AIR PATTERN	DAMPER	MATERIAL	FINISH	REMARKS
A	CEILING SUPPLY	4-WAY	OPPOSED DAMPER	ALUMINUM	OFF-WHITE	PERFORATED
B	SIDEWALL SUPPLY	4-WAY	OPPOSED DAMPER	ALUMINUM	OFF-WHITE	FRONT BLADE HORIZONTAL
C	CEILING RET/EXH	4-WAY	OPPOSED DAMPER	ALUMINUM	OFF-WHITE	EGG-CRATE
D	SIDEWALL EXHAUST	4-WAY	OPPOSED DAMPER	ALUMINUM	OFF-WHITE	FRONT BLADE HORIZONTAL

RECIPROCATING COMPRESSOR – CHILLER																	
UNIT NO.	NOM. (TONS)	EVAPORATOR DATA					COMPRESSOR DATA				SYSTEM KW/TON (MAX)	OPERATION DOWN TO °F AMBIENT	V	PH	HZ	COMPANION CONDENSER NO.	REMARKS
		GPM	FLUID	EWT (°F)	LWT (°F)	ΔP FT. H ₂ O	QTY	STAGES OF UNLOADING TOTAL	RATING (EA)	COMP. KW/TON (MAX)							
CH- 1&2	63.2	151.5	CH WTR	54.0	44.0	7.2	3	6	68.5	1.08	1.22	ZERO	460	3	60	CU-1&2	x

PUMP SCHEDULE											
SYMBOL	SERVICE	DATA							TYPE	REMARKS	
		GPM	TDH	RPM	EFF.	HP	VOLTS/PH	FLUID			
P-1	HOT WATER (MAIN CIRC.)	190	93	1750	68	10	460/3	HW	END SUCTION CENTRIFUGAL	VARIABLE FREQUENCY DRIVE	
P-2	HOT WATER(STANDBY)	190	93	1750	68	10	460/3	HW	END SUCTION CENTRIFUGAL	VARIABLE FREQUENCY DRIVE	
P-3	CHILLED WATER(CH-1)	150	93	1750	65	7 1/2	460/3	CHW	END SUCTION CENTRIFUGAL		
P-4	CHILLED WATER(CH-2)	150	93	1750	65	7 1/2	460/3	CHW	END SUCTION CENTRIFUGAL		

CABINET/UNIT HEATER SCHEDULE											
SYMBOL	SERVICE	TYPE	DATA					MOTOR			REMARKS
			CFM	TOTAL MBH	EWT (°F)	LWT (°F)	GPM	HP	SPD	RPM	
CUH-1	STAIRS	WALL RECESSED	265	48.0	160	130	2.0	65 WATTS	LOW	740	
CUH-2	VESTIBULE	WALL SEMI-RECESSED	270	9.3	160	130	2.0	65 WATTS	LOW	740	
CUH-3	DELIVERY	CEILING HIDEAWAY WITH PLENUM	135	12.7	160	130	1.0	35 WATTS	MED	840	
CUH-4	KITCHEN	CEILING HIDEAWAY WITH PLENUM	265	9.7	160	130	0.5	30 WATTS	LOW	655	
UH-1 & 2	MECHANICAL RM.	HORIZONTAL PROPELLER	210	4.9	160	130	0.5	9 WATTS	LOW	1350	
UH-3 & 4	ATTIC SPACE	HORIZONTAL PROPELLER	210	4.9	160	130	0.5	9 WATTS	LOW	1350	
UH-5 & 6	BULK STORAGE HOUSEKEEPING	HORIZONTAL PROPELLER	210	4.9	160	130	0.5	9 WATTS	LOW	1350	
UH-7	FIRE PUMP ROOM	HORIZONTAL PROPELLER	210	4.9	160	130	0.5	9 WATTS	LOW	1350	

FINNED TUBE RADIATION										
SYMBOL	LOCATION	ACTIVE LENGTH (FEET)	CAPACITY (BTUH/LF)	G.P.M.	NUMBER ROWS	FINS PER FOOT	TUBE SIZE	AVG. WATER TEMP. °F	ENCLOSURE	REMARKS
FT-1	AS-NOTED	AS-NOTED	500	3	1	40	3/4	150	SLOPE TOP/ LOW PROFILE	FIN SIZE: 3 1/4" x 3 1/4"
FT-2	AS-NOTED	AS-NOTED	1250	3	2	40	3/4	150	SLOPE TOP/ 24" HIGH	FIN SIZE: 4 1/4" x 3 5/8"

HEAT EXCHANGER									
SYMBOL	CAPACITY (MBH)	LIQUID	FLOW (G.P.M.)	LIQUID TEMP. °F		PRESSURE DROP (FT. OF H ₂ O)	STEAM PRESSURE (P.S.I.G.)	STEAM RATE (LBS./HR.)	REMARKS
				ENT.	LVG.				
HE-1	925	HOT WATER	190	150	160	2.0	5	931	10"ø x 24" LG. 2-PASS

CONDENSATE PUMPS									
SYMBOL	SERVES	CAPACITY (LB/HR)	DISCH. PRESS. (P.S.I.G.)	RECEIVER CAPACITY (GALLONS)	PUMP GPM	MOTOR			REMARKS
						QTY	HP	RPM	
CP-1	AHU-1,2 & 3	1500	25	14	9	2	1/2	3500	DUPLEX UNIT
CP-2	HE-1 & WTR HTRS	5000	50	36	30	2	2	3500	DUPLEX UNIT

graphic scales			ARROWSTREET INC SOMERVILLE, MA		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND				
CHECK GRAPHIC SCALES BEFORE USING			NORTHERN DIVISION		LESTER				
			BACHELOR ENLISTED QUARTERS		PENNSYLVANIA				
			EQUIPMENT SCHEDULES SHT.1		GROTON, CT				
		SIZE		CODE IDENT. NO.					
		F		P-422		NAVFAC DRAWING NO.		DIS. SH. NO.	
						2191638		M-11	
						CONSTR. CONTR. NO. N62472-96-C-0420			
						SPEC. 04-96-0420		SHEET 90 OF 135	

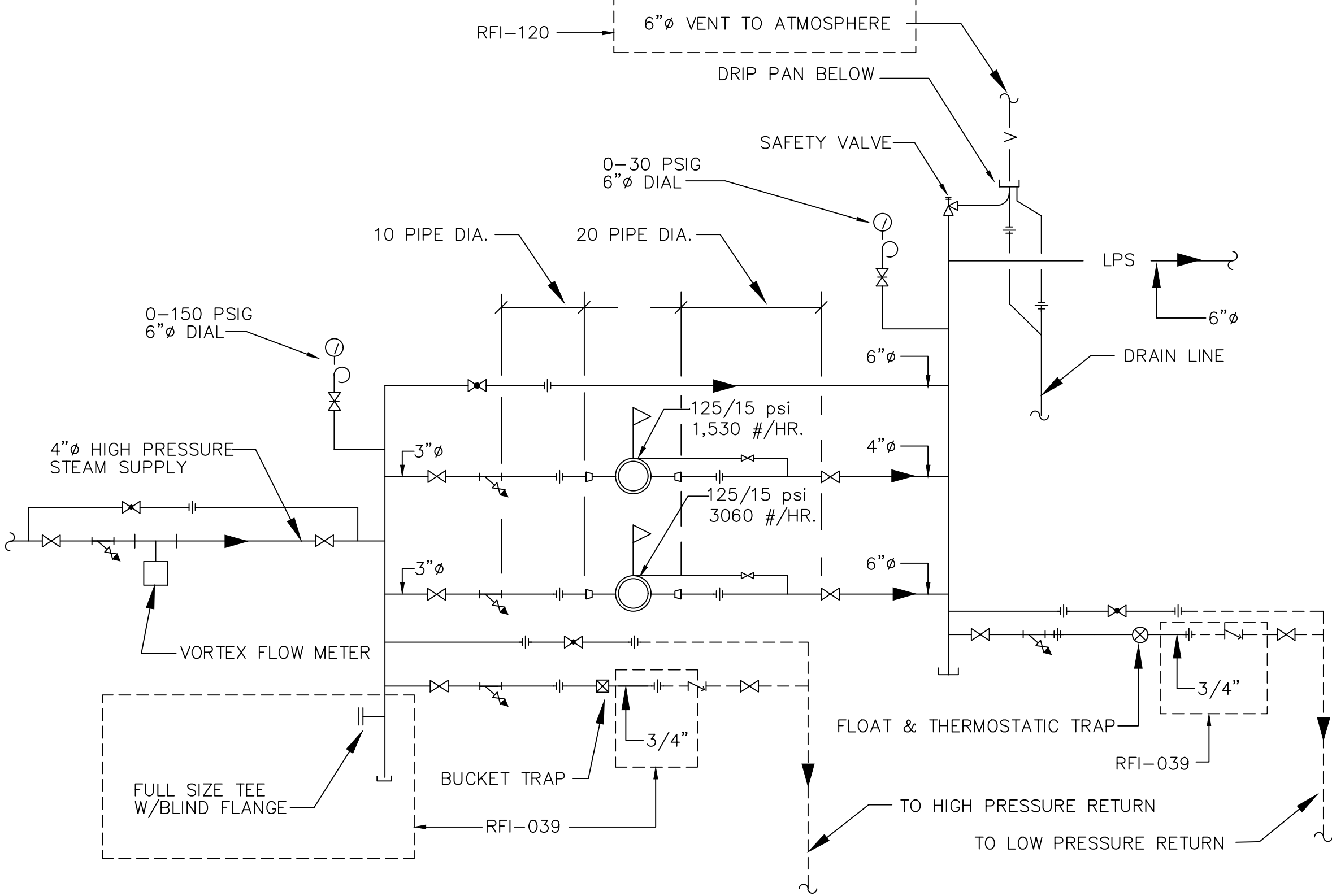
REVISIONS				
LTR	DESCRIPTION	PREP'D BY	DATE	APPROVED
	CHANGED TO AS - BUILT	J.P.	9/24/99	R.R.

AIR COOLED CONDENSING UNIT								
SYMBOL	SERVE	TOTAL HEAT REJECTION (MBH)	AMBIENT TEMPERATURE (°F)	ELECTRICAL		FAN MOTOR HP	CABINET SIZE LXWXH	REMARKS
				FLA	V/ø/HZ			
CU-1	CH-1	970.8	95.0	8.8	460/3/60	(3) @ 1	153.25" x 63.75" x51.75"H.	
CU-2	CH-2	970.8	95.0	8.8	460/3/60	(3) @ 1	153.25" x 63.75" x51.75"H.	

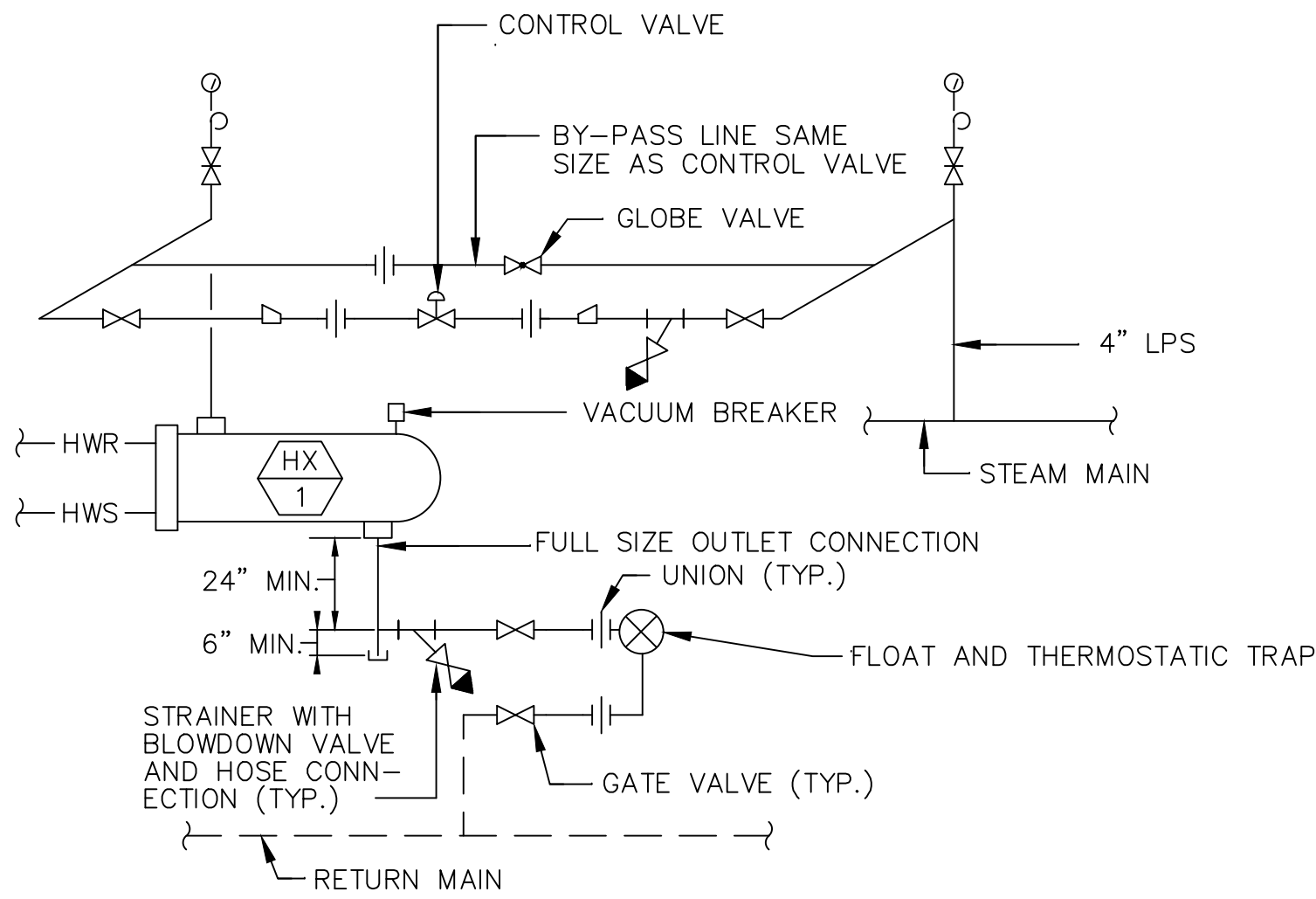
EXHAUST FANS												
SYMBOL	LOCATION	SERVICE	CFM	STATIC PRESS. (IN.WG)	H.P.	RPM	MAX. T.S./O.V. FPM	DRIVE	ELECTRICAL			REMARKS
									V	PH	CY	
EF-1	ROOF	TOILETS	420	0.25	1/4	960	3267	BELT	120	1	60	CENTRIFUGAL/UPBLAST
EF-2	ROOF	TOILETS	420	0.25	1/4	960	3267	BELT	120	1	60	CENTRIFUGAL/UPBLAST
EF-3	LEVEL 8	TOILETS	6300	0.75	1 1/2	695	1500	BELT	460	3	60	CENTRIFUGAL, UTILITY SET
EF-4	LEVEL 2-7	ELEC. RM.	730	1.00	1/2	1450	950	BELT	460	3	60	INLINE CABINET FAN
EF-5	LEVEL 8	LEVEL 8	4000	0.200	1/2	1050	5498	BELT	460	3	60	PROPELLER
EF-6	LEVEL 8	ELEVATOR MACH. RM.	2500	0.25	1/2	722	4542	BELT	460	3	60	PROPELLER
EF-7	GROUND FLR	TOILETS	500	0.75	3/2 W	1550	—	—	120	1	60	INLINE CABINET FAN
EF-8	MECH. RM.	MECH. RM.	3500	0.250	3/4	405	3800	BELT	460	3	60	PROPELLER
EF-9	MECH. RM.	MECH. RM.	3500	0.250	3/4	405	3800	BELT	460	3	60	PROPELLER
EF-10	ROOF	STAIRS	2000	0.375	1/2	600	3000	BELT	460	3	60	CENTRIFUGAL
EF-11	ROOF	STAIRS	2000	0.375	1/2	600	3000	BELT	460	3	60	CENTRIFUGAL
EF-12	LAUNDRY RM.	DRYERS	3424	0.375	3/4	2000	—	BELT	460	3	60	18"ø TUBE AXIAL
EF-13	LAUNDRY RM.	STAIRS	3424	0.375	3/4	2000	—	BELT	460	3	60	18"ø TUBE AXIAL
EF-14	KITCHEN	MEETING RM.	1500	0.500	1/2	920	760	BELT	460	3	60	INLINE CABINET FAN
EF-15	GROUND FL.	ELEC. RM.	1320	1.00	3/4	1343	1100	BELT	460	3	60	INLINE CABINET FAN
EF-16	MECH. RM.	EMERGENCY EXHAUST	1200	0.250	1/6	1725	6327	DIRECT	120	1	60	PROPELLER
SF-1	FIRE PUMP ROOM	FIRE PUMP ROOM	2590	0.250	1/4	1140	5974	DIRECT	120	1	60	PROPELLER (ARRANGED FOR SUPPLY)

* NOTE: THERE IS A UNIT EF-4 PER EACH OF 6 ELEC. RMS.

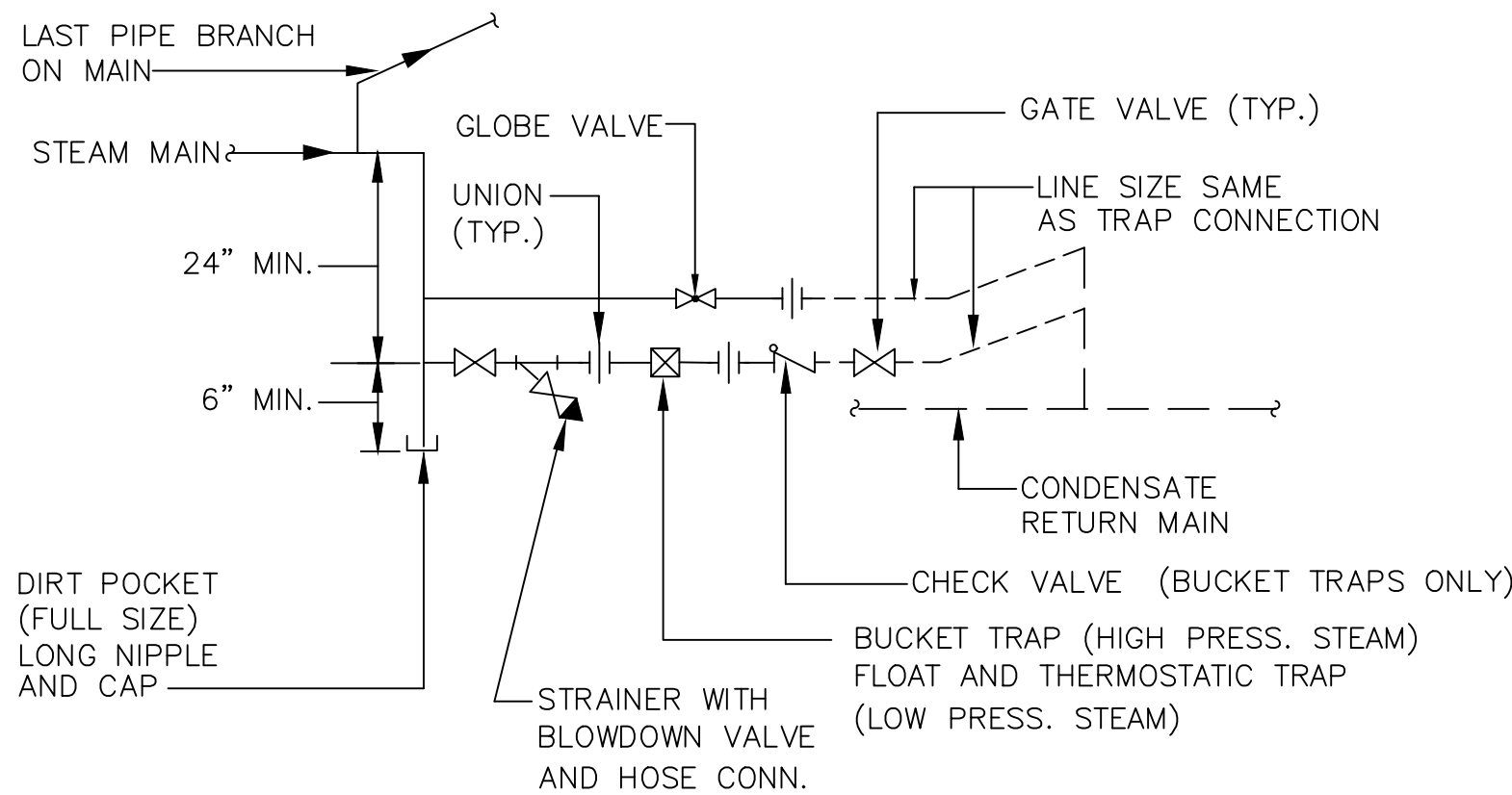
SOUND ATTENUATORS					
SYMBOL	SERVICE	LOCATION	CAPACITY (CFM)	PRESS. DROP (IN. W.G.)	REMARKS
SA-1	MEETING ROOM	MECH.ROOM	900	0.05	24"x12"x5'-0"LG.
SA-2	MEETING ROOM	MECH.ROOM	2400	0.10	24"x24"x5'-0"LG.
SA-3	LAUNDRY ROOM	MECH.ROOM	6848	0.06	60"x24"x5'-0"LG.
SA-4	QUARTERS	LEVEL 8	9400	0.06	48"x24"x5'-0"LG.
SA-5	QUARTERS	LEVEL 8	1200	0.06	24"x24"x5'-0"LG.



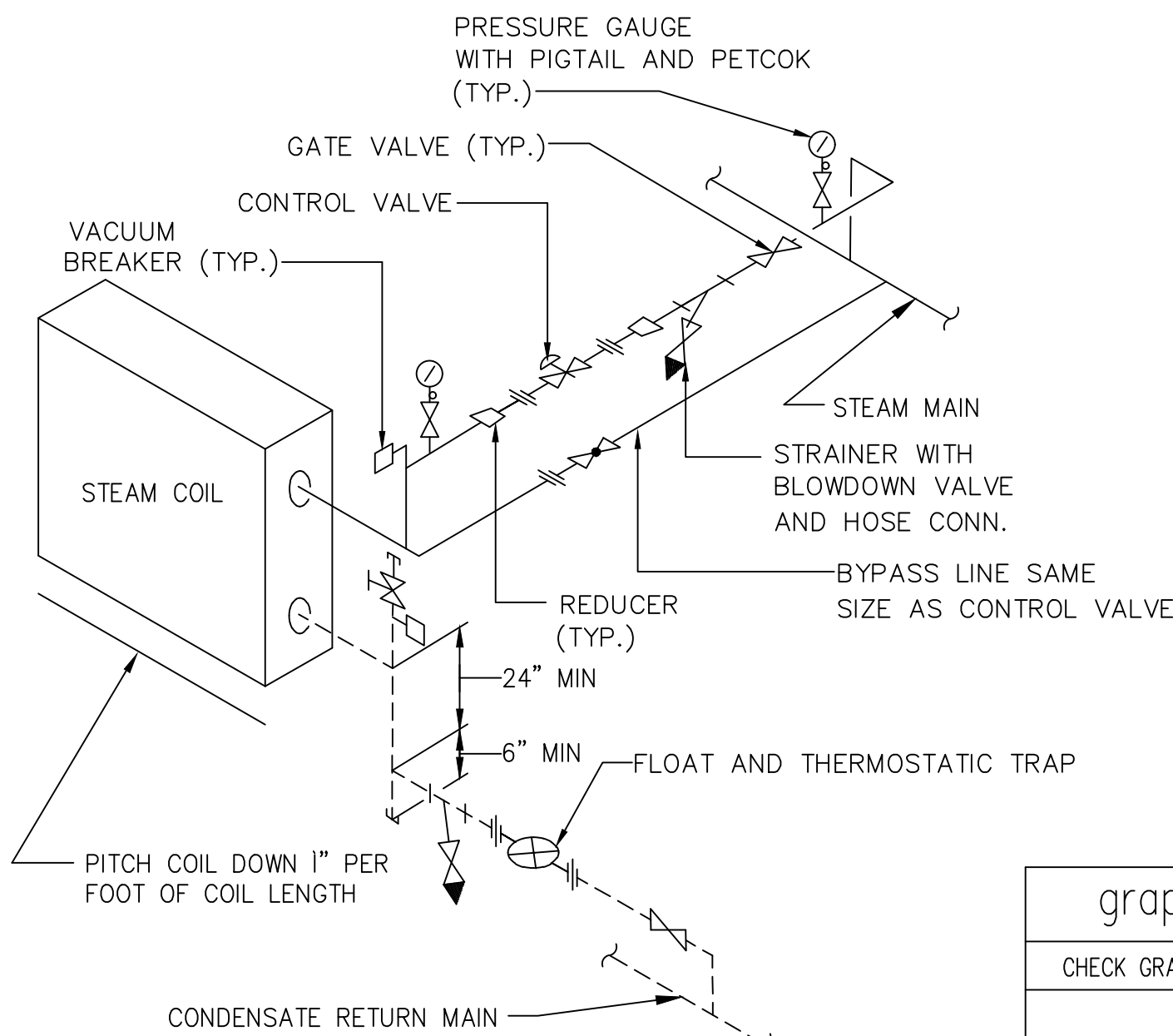
STEAM PRESSURE REDUCING VALVE STATION PIPING DETAIL
NO SCALE



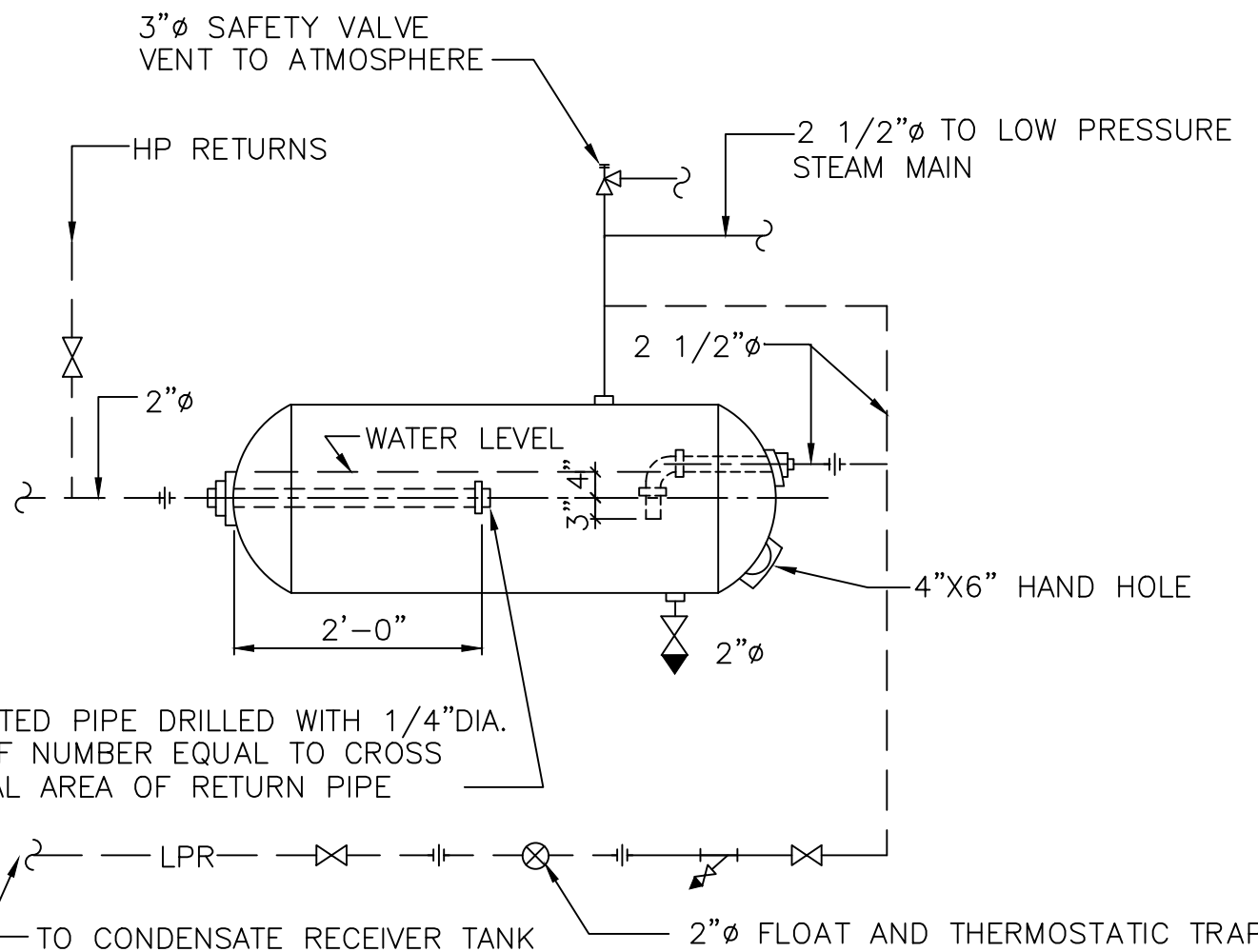
HEAT EXCHANGER STEAM PIPING DETAIL
NO SCALE



END OF STEAM MAIN AND RISER
DRIP DETAIL
NO SCALE

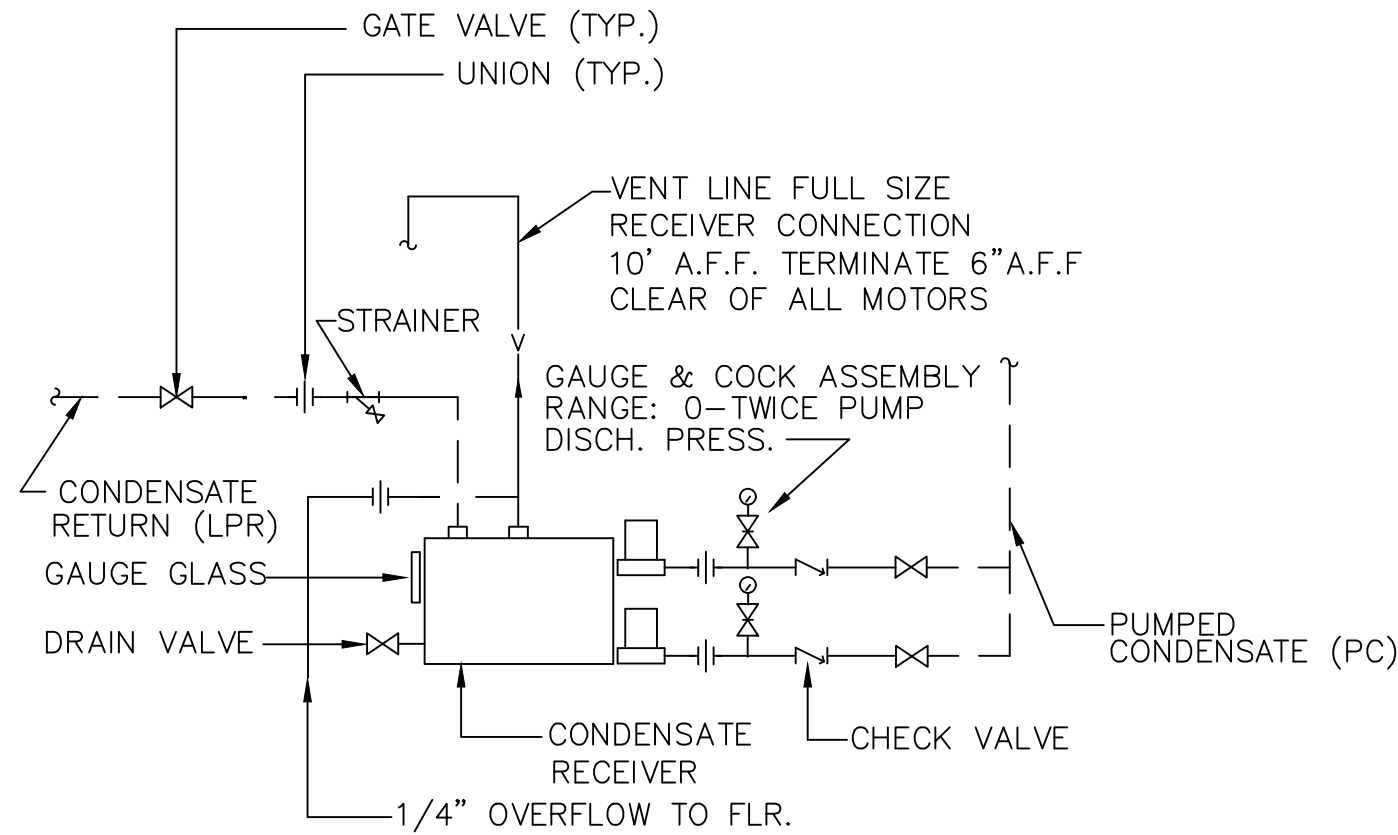


SINGLE SECTION STEAM COIL
NO SCALE



FLASH TANK SCHEDULE			
LOCATION	LB. STM/HR.	DIA.	LENGTH
MECH. ROOM	413	1'-0"	4'-6"

FLASH TANK DETAIL
NO SCALE



CONDENSATE PUMPING UNIT PIPING DETAIL
NO SCALE

graphic scales		ARROWSTREET INC SOMERVILLE, MA		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND NORTHERN DIVISION LESTER NAVAL SUBMARINE BASE-NEW LONDON PENNSYLVANIA GROTON, CT	
CHECK GRAPHIC SCALES BEFORE USING		DSGN R.D.L. DR R.D.L. CHK R.R. SUPV R.R. CHK ENGR R.R. SUBMITTED BY: W. DANIEL / PRINCIPAL DATE: 28 MARCH 1997 FIRM MEMBER (TITLE) NORTHDIV HD DM HD PPE DR		SIZE CODE IDENT. NO. NAVFAC DRAWING NO. DIS. SH. NO. F P-422 2191639 M-12 CONSTR. CONTR. NO. N62472-96-C-0420 SPEC. 04-96-0420 SHEET 91 OF 135	
SATISFACTORY TO DATE		APPROVED IN CHARGE DATE		APPROVED DATE	
NORTHDIV FOR COMMANDER NAVFAC		SCALE NONE		SHEET 91 OF 135	